

The Right to Health and the Health Effects of Denials*

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Abstract

We investigate supply-side barriers to medical care in Colombia, where citizens have a constitutional right to health, but insurance companies impose restrictions. We use administrative data on judicial claims for health as a proxy for unmet demand. We validate this using the health services utilization register, showing that judicial claims map into large, pervasive decreases in medical consultations, procedures, hospitalizations and emergency care. This manifests in population health outcomes. We identify increases in mortality pervasive across cause, age, sex, and income, with particularly pronounced effects for certain cancers.

JEL classifications: G22, I11, I13, I18, K38, K42

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1 Introduction

Universal health coverage (UHC) has, by now, garnered widespread support as a policy objective (Bloom et al. 2018). It is seen as marking the third great transition in public health, following the demographic and epidemiological transitions of the past two centuries (Rodin and Ferranti 2012). Evaluations of UHC commonly look at measures of coverage, access, and financial risk protection (O’Connell et al. 2014; Reich et al. 2016; WHO 2023), yet evidence suggests that expanding coverage does not always lead to better population health outcomes. Some studies highlight low uptake (Remler et al. 2001; Currie 2004; Aizer 2007; Banerjee et al. 2021; Ericson et al. 2023; Banerjee et al. 2024), while others highlight poor service quality (Barber et al. 2017; Kruk et al. 2018; Banerjee et al. 2024). We examine access restrictions driven by the financial incentives of private insurers to deny or delay medical claims within a managed care system that has nominally achieved UHC. Using a novel measure derived from a database of legal claims against health insurers, we estimate the impact of denials on the utilization of medical services and population-level health outcomes.

The setting is Colombia, where a commitment to the right to health for all citizens was written into the Constitution of 1991, following which a new healthcare system that progressively reached nearly universal health coverage was put in place. At 99 percent, Colombia has one of the highest levels of comprehensive insurance coverage among developing economies. While over half of UN member states constitutionally recognize the right to health (Kinney and Clark 2004; Backman et al. 2008; Ruger et al. 2015), Colombia stands out by providing robust judicial protection. The Colombian Constitutional Court has ruled that obstacles to accessing health services imposed by health insurers breach the right to health.¹ Citizens can directly petition local courts if they believe their health rights are violated, creating a unique accountability mechanism. These legal claims, known as *tutelas*, are cost-free and easy to file. *Tutelas* are given preferential treatment, with judges required to issue a decision within ten days.²

The use of *tutelas* as a legal enforcement mechanism has become widespread in the country, contributing to what is commonly referred to as the judicialization of health care (Yamin and Parra-Vera 2009, 2010; Vargas et al. 2010; Parra-Vera and Yamin 2013; Andia and Lamprea 2019). Between 2010 and 2016, citizens filed approximately 90,000 *tutelas* annually against health insurers. A key feature of our study is that we use administrative

1. Colombia is one of 63 countries that has an independent constitutional court, a high court that deals primarily with constitutional law. Other countries, including the US, delegate constitutional judicial authority to their general court system, with the final decision-making power resting in the Supreme Court.

2. Similar judicial protection mechanisms exist in Argentina (Bergallo 2011), Brazil (Biehl et al. 2009; Martins and Allen 2020), Costa Rica (Monge 2019) and South Africa (Wilson 2011), though their use is much less widespread.

register data on these judicial claims to quantify the prevalence of supply-side barriers to medical care. We then analyze how these unmet healthcare needs—represented by services demanded but not delivered—affect mortality rates, both overall and for specific causes of death.

Restrictions on access to medical care imposed by insurers reflect the financial incentives inherent in the design of Colombia’s healthcare system. Since 1993, Colombia has pursued universal health coverage through a managed competition model of health insurance. Under this model, insurers must provide a standardized, comprehensive benefit package defined by the government. In exchange, they receive fixed capitation payments and are prohibited from engaging in risk-based selection of enrollees. As a result, the strategic options typically available to insurers are heavily regulated. The combination of fixed capitation payments and variable costs tied to service utilization creates strong incentives for insurers to limit access as a cost-containment strategy. These incentives are reinforced by a weak enforcement regime, which reflects the limited administrative capacity commonly observed in many low- and middle-income countries (Banerjee et al. 2021; Banerjee et al. 2024). While this institutional arrangement is specific to Colombia, major insurers in the United States—such as UnitedHealthcare, Humana, and CVS—face similar allegations (Armstrong 2023; Ross and Herman 2023; Abelson 2024), particularly in the context of Medicare Advantage Plans (US Senate 2024).

To motivate our empirical strategy, we sketch a model that captures the key determinants influencing insurer-level decisions to delay or deny health claims, or to impose strict conditionalities, and citizen-level decisions to litigate when faced with these restrictions. Denial rates and the proportion of denials resulting in litigation are equilibrium outcomes that jointly determine the tutela rate (i.e., the number of tutelas filed against health insurers per 1,000 enrollees). These outcomes depend on both demand-side factors (e.g., population health status, probability of winning the case, and the non-pecuniary costs of litigation) and supply-side factors (e.g., insurer-specific costs of providing services and the administrative costs associated with handling claims). We show that the tutela rate (which is observed) is a scalar multiple of unmet demand (which is unobserved), creating a strong link between variations in the two. The model suggests that some restrictions may be excessive while others may be optimal from a cost-benefit perspective, particularly given the rise in healthcare spending, a challenge common to many systems (Chernew and Newhouse 2011; OECD 2015). In fact, the extensive protection of the right to health allows individuals to file claims

even for services with potentially limited health impacts.³

We obtained data on all judicial claims (*tutelas*) by insurer, municipality, and year through a petition filed under a Right to Information Act. We combined this with administrative data indicating which insurer every citizen was enrolled with in each municipality and year; a register containing the universe of medical services provided, classified by type of service (consultations, procedures, hospitalization, emergencies); and with vital statistics data that contain individual level mortality (a widely used marker of population health) that we collapse to the insurer-municipality-year level.

The main challenge to identification is isolating variation in *tutela* rates driven by supply-side restrictions rather than demand-side factors. By virtue of using insurer-municipality-year data, we can control for cross-sectional heterogeneity at the insurer-municipality level and for shocks at the insurer-year and at the municipality-year level.⁴ To address the concern that omitted variables varying at the insurer-municipality-year level may still bias our results, we adopt an instrumental variables approach. This strategy leverages the fact that insurers operate across multiple municipalities, allowing us to isolate insurer-specific (supply-side) factors influencing restriction rates that are orthogonal to local demand for health care.⁵ Specifically, we construct an instrument based on the insurer-specific *tutela* rate, calculated using data exclusively from municipalities outside the department where the index municipality is located but where the insurer operates.⁶

Another potential threat to our identification strategy is dynamic selection. If individuals switch insurers in response to restrictions, this could create selectivity or endogenous heterogeneity in the risk profiles of enrollees across insurers. To address this concern, we model enrollee mobility across insurers using individual-level data and demonstrate that switching behavior is unrelated to the number of judicial claims filed in the year prior to the change.

We investigate the presumed mechanism by examining the impact of variation in *tutela* rates on the utilization of medical services. If health shocks were driving changes in *tutela* rates, we would expect a positive association between *tutela* rates and service utilization. Conversely, if supply-side restrictions are the primary driver, these restrictions should reduce

3. We recognize that there is an additional relevant decision, which is whether the judge grants the claim. In our analysis, variation in judge leniency is purged with fixed effects for insurer-municipality and municipality-year.

4. These are fixed effects for each pair of the three dimensions of the data cell. Between them these controls capture not only relevant baseline differences but also shocks to health, including infections, demographics, information, the efficiency of provision, system-level changes, cultures of claiming, and the leniency of local courts.

5. Natural examples of insurer-specific supply-side factors are cost structures determined by different service provider contracts and managerial practices.

6. Section 6 explains the plausibility of the exclusion restriction conditional on the three-way fixed effects.

service utilization. Our findings indicate that restrictions significantly decrease the utilization of medical services across the board. An increase of one tutela per 1,000 enrollees—equivalent to nearly a 50 percent rise in the tutela rate—is associated with a 13.6 percent decline in consultations, a 35.7 percent decline in procedures, a 10.9 percent decline in hospitalizations, and a 36.4 percent decline in emergency services, all relative to their respective means.

We find large effects of supply-side restrictions on mortality rates. Using our preferred IV specification, we estimate that an additional tutela per 1,000 enrollees is associated with an increase of between 7.6 and 11.6 percent in the mortality rate relative to the mean. These estimates are broadly similar whether or not we use an instrument, increasing our confidence that selection is not a major issue.

We find that the impact of restrictions on mortality is pervasive, manifesting across causes of death, age groups, sexes, and broad income categories. For instance, we find large effects on mortality rates due to certain cancers, aligning with the fact that neoplasms are among the most commonly cited diagnoses in tutela claims. Mortality rates are widely used as a key indicator of population health (Case and Deaton 2017), given their consistent availability over extended periods in vital statistics data. However, since mortality is an extreme measure that represents the tip of the iceberg of underlying morbidities, we might regard our estimates as a lower bound.

While there is undeniable value to lives lost, one might hypothesize that insurers are more likely to deny treatment to an individual who would die irrespective of treatment. Gruber (2022) highlights that 21 percent of health care spending in the United States is concentrated among just one percent of the population, while nearly one-quarter of Medicare spending occurs in the last year of life (Einav et al. 2018). This raises important questions about cost-efficiency and the potential for an optimal rate of restrictions. Although we cannot fully address this issue, we assess it using the age-adjusted Charlson Comorbidity Index, a widely adopted tool in medical practice for predicting ten-year mortality in patients with co-morbid conditions. We can reject that restrictions are targeted toward individuals with a higher latent risk of mortality, instead, we see an increase in mortality across the distribution of the risk index. This aligns with our theoretical framework, which posits that medical care with both low and high expected benefits is likely to face restrictions. Moreover, given the inherent unpredictability of death, this raises further questions about the feasibility of ex-ante resource prioritization. As emphasized by Einav et al. (2018), those who end up dying are not necessarily the same as those who were sure to die.

We investigated whether restrictions are greater in municipalities with weaker insurer-level competition and found no significant association. We similarly find no evidence that increased mortality emerges from municipalities where insurers face lower competition. This

is consistent with the literature on the industrial organization of healthcare markets which shows, for the American market, that the impact of insurer competition on welfare, negotiated provider prices, and premia is theoretically ambiguous (Ho and Lee 2017). In our setting too, it is unclear that enforcing more competition will induce better provision, for instance because greater market power may enhance the bargaining power of insurers when negotiating contracts with providers, generating cost reductions that lower the incentive to impose restrictions on demand.

With the same Right to Information petition that we used to obtain judicial claims data, we also obtained data on the universe of complaints against health insurers made to the National Health Superintendent (Supersalud), the agency in charge of oversight of the system. This is an alternative measure of supply-side restrictions from an independent source. We consistently show that our estimates are similar using complaints. This increases robustness of our findings to concerns over measurement.

The recent assassination of UnitedHealthcare’s CEO in the U.S. has brought to prominence the deep-set dissatisfaction of patients (and, often, doctors and policymakers) with the delay, deny and defend tactics of private health insurance firms, documented more than a decade ago by Feinman (2010). This issue is also highly relevant in Colombia, where healthcare remains a central electoral concern. The rise of the left-wing party in 2022 occurred amid debates over managed care provision, and for the past three years, Congress has been embroiled in a contentious debate over healthcare reform, particularly the financial incentives shaping insurers’ decisions. Our point of departure in this paper is the incongruent juxtaposition of the constitutional commitment to health coverage with a managed health care model of provision in which insurers behave similarly, albeit under different constraints that strengthen their incentive to deny and delay approvals for health diagnoses and treatments (Maya 2008; Corte Constitucional 2008; Yamin and Parra-Vera 2009, 2010; Parra-Vera and Yamin 2013; Arrieta-Gómez 2018).⁷ Citizens have responded to these restrictions by pursuing litigation. Our results indicate that litigation – afforded by the state being accountable for its commitment to citizen rights to health - can be effective in fighting arbitrary restriction, but it is not enough.

Our results highlight the costs to population health associated with the incentives of insurers being unaligned with the commitment to universal health coverage. There may be too much regulation insofar as insurers are not allowed to set prices or select enrollees and, at the same time, too little regulation insofar as insurers are not sufficiently penalized when complaints against them are received and upheld.

7. See also the periodical reports of the Colombian Ombudsman office (Defensoría del Pueblo 2004, 2007, 2009, 2010, 2011, 2012, 2013, 2015); and reports by the Ministry of Health (Minsalud 2015).

The rest of this paper unfolds as follows. Section 2 describes the related literature, demarcating our contributions. Section 3 outlines the relevant features of the Colombian health care system and the process of judicialization of access to health care. Section 4 describes the data sources. Section 5 discusses our theoretical framework. The econometric model and identification strategy are presented in Section 6. In Sections 7 and 8 we present our main results, while Section 9 examines the targeting of restrictions based on comorbid conditions. Finally, Section 10 concludes.

2 Related Literature

Our contribution is that we look at a system with a constitutional commitment to universal health insurance coverage, and highlight the impact of restrictions on access to health care imposed by private insurers. A unique feature of our approach is that we exploit administrative data on judicial claims, demonstrating (with full population data on all medical service utilization) that they carry information on barriers to accessing prescribed medical treatments including delays, denials and conditionalities. In general, there are no systematic at-scale data that measure such barriers. We sketch a model of the key decisions that are involved- the decision of the insurer to impose a restriction, and the decision of the citizen to litigate in response. The model formalizes the notion that information on unmet demand is captured in judicial claim volumes, and it illuminates the identifying assumptions that support our empirical strategy.

In this section we discuss how our work brings new evidence to strands of a broader literature studying litigation in healthcare, the expansion of health coverage, the incentives of medical providers, and the inefficiencies caused by administrative hurdles.

There is a literature on the impact of litigation on the the delivery of healthcare services that has predominantly focused on the impact of medical malpractice liability and tort reform, showing nuanced findings regarding healthcare delivery and outcomes (Kessler and McClellan 1996, 2002; Currie and MacLeod 2008; Frakes 2013; Frakes and Jena 2016). The evidence suggests that while malpractice liability may incentivize quality improvements in specific contexts, it also promotes costly defensive medicine without clear health benefits. Tort reforms that reduce liability exposure appear to achieve cost savings without adverse health effects, though impacts vary across medical specialties and jurisdictions. As Danzon (2000) notes in a comprehensive review, while litigation costs are substantial, the deterrent effects of malpractice law on physician behavior remain uncertain.

We do not (cannot) analyze the impacts of judicial accountability for the right to health. Instead, we exploit that judicial claims carry information on denials of prescribed medical

treatments to analyse impacts on healthcare utilization and health outcomes. Our setting draws attention to constitutional design and accountability issues, highlighting a lacuna in modeling the constitutionalization of citizen rights. An explicit commitment to fulfilling rights such as the right to health or education or food increases salience and, when supported by the powers of a constitutional court, provides a compelling framework for enacting human rights law. We are unaware of research that evaluates this framework’s incentive structure or welfare consequences. In broadly related work, Fox and Stephenson (2011) model judicial review, analyzing its impacts on the incentives of elected leaders to posture by enacting bold but possibly ill-advised policies and on voter welfare. Maskin and Tirole (2004) consider the implications of constitutional design for public choices, analyzing the strengths and weaknesses of electoral accountability and the role of non-accountable officials, including judges.

Our paper is related to the extensive literature studying the impact of access to health insurance on the utilization of health services and population-level health outcomes. For developed countries, particularly the United States, this literature generally finds that extending health insurance has positive impacts on health outcomes and mortality. For instance, early Medicaid implementation reduced infant and child mortality significantly (Goodman-Bacon 2018, 2021), with benefits persisting up to 50 years later. More recent evidence from the Affordable Care Act Medicaid expansions shows a decline in annual mortality (Miller et al. 2021). While the Oregon Health Insurance Experiment found no significant improvements in measured physical health markers, it demonstrated reduced financial strain and increased healthcare utilization (Baicker et al. 2013). Other studies of U.S. insurance expansions consistently show mortality reductions, particularly among vulnerable populations such as children, older adults, and minority groups (Currie and Gruber 1996; Card et al. 2008; Sommers et al. 2012).

Evidence from low- and middle-income countries presents more mixed results. While most experimental studies in these settings find limited impacts on health outcomes (Banerjee et al. 2024), there are notable exceptions. China’s New Cooperative Medical Scheme significantly lowered mortality rates, reportedly saving over one million lives annually at its peak (Gruber et al. 2023). Studies from Colombia and Taiwan also demonstrate positive impacts: Colombia’s health reform improved self-reported health and healthcare utilization (Gaviria et al. 2007), while Taiwan’s National Health Insurance significantly reduced infant mortality among farm households (Chou et al. 2014). We contribute to this literature by identifying the extent to which slippage between coverage and access can hamper the realization of the potential benefits of coverage. This is important as it indicates scope for policymakers to consider design issues that could increase the impact of public funds. While

the cited studies primarily look at impacts of expanding health insurance coverage, we focus on the manner in which the financial incentives of private insurers can limit the benefits of expansion.

Thus, our paper also contributes to the literature on how financial incentives influence healthcare provision. Evidence suggests that higher reimbursement rates lead to increased service provision and improved access to care (Currie et al. 1994; Polsky et al. 2015; Banerjee et al. 2024). The evidence from major policy changes, such as Medicare’s introduction (Finkelstein 2007), demonstrates that supply-side responses like hospital expansion and technology adoption can have system-wide effects that extend beyond the directly targeted population. This suggests that effective healthcare reform requires careful attention to provider incentives and market structure, not just insurance coverage expansion. Likewise, cost-control incentives in managed care settings can reduce unnecessary care and expenditures (Gaynor et al. 2004). However, they may also lead to unintended consequences such as patient selection (Alexander 2020) or reduced access for certain populations (Duggan et al. 2023). We depart from this literature by focusing on financial incentives of health insurers. We show that if there is weak enforcement within managed care models, cost-control incentives can end up limiting access to necessary services with strong impacts on population health.

Another strand of the literature highlights that administrative burdens and transaction costs create significant barriers to both healthcare access and insurance utilization, particularly in public insurance programs. Evidence from both developed and developing countries shows that administrative frictions substantially reduce program effectiveness (Banerjee et al. 2021; Malani et al. 2021; Dunn et al. 2024). However, the literature also suggests that well-designed administrative systems can effectively improve healthcare delivery while controlling costs. For example, Medicare’s Recovery Audit Contractor program generates substantial returns, with every \$1 spent on audits recovering \$24-\$29 in savings without apparent negative health impacts (Shi 2024). Studies of prior authorization in Medicare Part D show that such restrictions can reduce unnecessary drug utilization without causing significant adverse health outcomes (Brot-Goldberg et al. 2023). Conversely, removing overly restrictive administrative barriers, such as prescription drug caps in Medicaid, can improve health outcomes and reduce costly hospitalizations (Layton et al. 2022). We contribute to this literature by showing that unsupervised—and in many cases illegal—administrative barriers can restrict access to services included in benefit packages, with significant health consequences. While administrative processes are necessary for program integrity, they must be carefully designed to balance cost control with healthcare access.

3 Institutional Background and Policy Context

Structure. The Colombian health care system operates under a tightly regulated managed competition model for health insurance. Citizens enter the system by enrolling with a health insurer (Entidad Promotora de Salud, EPS) that operates within their municipality of residence. These insurers bear responsibility for ensuring their enrollees have access to essential healthcare services offered through a network of independently contracted service providers, including hospitals, clinics, and laboratories. Insurers establish contracts with public and private healthcare providers, fostering competition in service delivery. Citizens can freely select their health insurer and the providers within that insurer’s network.

Two regimes operate under a similar structure but target different populations: the Contributory Regime (CR) covers formal sector employees and self-employed individuals with the capacity to pay. The Subsidized Regime (SR) targets low-income individuals, with eligibility determined through means testing. The high coverage of the system is a marker of success. The share of the population enrolled with an insurer in either the CR or the SR between 2010 and 2018 was 90 percent on average (Appendix Figure A.1), with half the covered population in each regime. By 2021, this coverage had expanded to 99% of the population.

Insurers face regulatory constraints on three key strategic variables. First, insurers must provide a standardized mandatory benefit package (Plan Obligatorio de Salud, POS) that encompasses a comprehensive range of healthcare services, including preventive and health promotion services, outpatient and inpatient care, surgical procedures, medications and treatments, rehabilitation and palliative care, mental health services, dental care, maternal and child health, chronic disease management, as well as transplants and specialized care. The Ministry of Health defines and periodically updates the composition of the POS to ensure it remains comprehensive and adequately addresses typical healthcare needs. Consequently, the basic healthcare plan is uniform across the population.⁸

Second, instead of setting differentiated prices for the POS, insurers receive a government-defined capitation payment, periodically recalibrated. Capitation rates vary by age, gender,

8. Enrollees in the CR have the option to purchase complementary insurance plans offered by their insurers. These supplementary plans, which operate outside the scope of the POS, provide enhanced levels of care and differentiated services. However, due to the comprehensive nature of the POS and the additional cost of these plans, only about 8% of the population chooses to purchase complementary coverage.

region, and regime but are uniform across insurers within these categories.^{9,10} Similar to Health Maintenance Organization (HMO) models, insurers bear the full financial risk of excess utilization of services covered under the POS—or the full benefit of limiting care (Gruber 2022). Users must pay out-of-pocket for services or medications excluded from the POS unless the treatment is deemed indispensable and unaffordable. In such cases, insurers are required to provide timely access, with the government reimbursing the costs.

Finally, insurers are prohibited from denying enrollment based on demographics or pre-existing conditions, preventing direct risk selection. However, limited competition among insurers within municipalities reduces consumer choice. Within the CR, the average municipality has 6.5 insurers to select from, while this number drops to 4.2 for the SR (Panel (a) of Appendix Figure A.2). Moreover, around 10% of municipalities have a single insurer in the SR (Panel (b) Appendix Figure A.2), while between 10% and 20% of municipalities offer at most two CR insurers (Panel (c) of Appendix Figure A.2).

Restrictions. Fixed capitation payments coupled with variable service costs generate strong financial incentives for health insurers to restrict healthcare access. Although certain access restrictions may be economically justified to mitigate demand-side moral hazard (Miller et al. 2013; Gruber 2022), insurers often implement arbitrary and potentially harmful barriers that primarily serve profit-maximization objectives rather than efficiency considerations (Feinman 2010). Extensive evidence from Colombia demonstrates that insurers systematically impose unauthorized barriers to limit healthcare utilization (Maya 2008; Abadia et al. 2009; Yamin and Parra-Vera 2009; Rodríguez 2012; Arrieta-Gómez 2018). These access restrictions manifest through multiple mechanisms: delays or denials in authorizing prescribed medications, examinations, and procedures; implementation of complex administrative processes; imposition of excessive documentation requirements; and creation of bureaucratic obstacles. The prevalence and persistence of these practices is exacerbated by inadequate regulatory enforcement (Maya 2008; Corte Constitucional 2008; Rodríguez 2012).

9. Capitation payments account for approximately 82% of total public health expenditures, which increased from 4.1% of GDP in 2011 to 4.7% in 2018. The remaining 18% is primarily allocated to two components: public health program expenditures (averaging 8.4%) and government-covered services outside the POS (averaging 7.4%). Funding for the system comes from multiple sources, with payroll contributions to the contributory regime—shared by workers and employers—representing 46% of total resources on average. The rest originates mainly from government allocations through the general budget (Granger et al. 2023).

10. There is a cost-sharing scheme through co-payments and moderating fees for outpatient care services designed to regulate healthcare utilization and promote appropriate service use. However, These cost-sharing mechanisms represent a minimal portion of total public health expenditures—averaging 1.61%—and primarily apply to higher-income individuals (Buitrago et al. 2023; Granger et al. 2023).

Tutela Writs. In response to systemic barriers imposed by insurers, Colombian citizens have increasingly resorted to judicial claims provided by Colombian law, notably the tutela writ. The Colombian Political Constitution of 1991 expressly enshrines a wide array of fundamental rights for all residents, including the right of any citizen to the “*highest attainable standard of health*”, which includes “*access to health services and technologies that guarantee comprehensive, timely, and high-quality care*”. Importantly, the Constitution introduced a legal enforcement mechanism known as the tutela writ. This writ is a cost-free legal petition that facilitates immediate access to the judiciary when an individual’s rights are perceived to have been violated. Its simplicity and efficiency characterize it: i) it is free of charge; ii) it can be filed with any judge in the relevant local jurisdiction; iii) it requires only the essential facts of the case for the judge’s review; and iv) it is treated as a priority, requiring the judge to issue a provisional ruling within ten days.

The Constitutional Court has explicitly stated that when insurers create obstacles to accessing health services, including those not covered by the mandatory benefit package, it constitutes a breach of the right to health.¹¹ An individual may file a tutela against an insurer for undue delays or denials of service, and a judge’s favorable ruling forces the insurer to grant the service. Typically, no additional penalty is imposed upon the insurer beyond the mandate to authorize the services requested, but regulatory agencies keep track of the number of tutelas and use this information to intervene if they are unusually high. Since some denials do not lead to tutela claims and some claims are not granted, this incentivizes the insurer to continue restricting access.

Two examples illustrate how people use tutelas to enforce access to medical care. In one case, a plaintiff acted on behalf of his five-year-old son, diagnosed with a brain tumor in June 2018. The health insurer authorized monthly chemotherapy covered by the POS, but the treatment was scheduled in a city over 380 kilometers away from their residence. The plaintiff requested a location change due to the inability to cover transportation costs, but the insurer did not respond. In August 2018, the plaintiff filed a tutela. The court ruled in favor of the plaintiff, ordering reassignment to a nearby provider and, if necessary, the provision of transportation. However, the treatment had already been delayed for over two months.

11. Section 3.2.1.3 of sentence T-760 of 2008 articulates: “*Constitutional jurisprudence has determined that access to a necessary health service, included in the mandatory plans, is a fundamental right in itself. Consequently, the denial of health services included in the POS infringes upon the fundamental right to health, making it a justiciable issue rectifiable by a Tutela writ.*” The Court further notes regarding services not included in the mandatory benefit package: “*when an individual needs a health service not covered by the Mandatory Benefit Plan and cannot afford it, the entities responsible for providing the service must uphold their duty and ensure access. In such cases, the government is responsible for covering the costs to guarantee the full exercise of the right to health.*” (Corte Constitucional 2008).

In a second case, the plaintiff acted on behalf of her mother, diagnosed with insulin-dependent diabetes. There was an urgent need for specialized consultation and two diagnostic imaging exams: an abdominal aortogram and peripheral arteriography. The health insurer denied the request, incorrectly stating that the POS did not cover these exams. After the denial, the plaintiff filed a tutela for urgent medical procedures. The court mandated immediate and continuous healthcare without barriers, but the patient had already lost a leg.

In both examples, insurers imposed illegal barriers to limit service utilization. Although required services were ultimately provided as a result of judicial rulings, the damage from the initial restrictions had already been done.

Due to its simplicity and fast turnaround, the tutela has become a very popular instrument in the country. The number of tutelas invoking *any* fundamental right increased from 0.3 to 12.7 per 1,000 inhabitants between 1992 and 2016 (See Panel (a) of Figure 1),¹² with tutela writs invoking the right to health accounting for 20 to 40 percent of all claims. The growth over time has been substantial: in 1999, the total number of tutelas presented to courts invoking the right to health was 21,301 (0.53 per 1,000 inhabitants), but by 2016 the number reached 163,977 (3.36 per 1,000 inhabitants), more than sevenfold increase in a span of 17 years.¹³

Four features of the data on tutelas invoking the right to health are noteworthy. First, a significant majority, ranging from 65.9% to 82.4%, target insurers (Panel (a) of Figure 2). Less than 4 percent of tutelas are made against health service providers, and between 15 and 30 percent are made against local authorities, governmental agencies, or other actors in the system. Second, over 65% of the services petitioned in tutelas were included in the mandatory benefits package (Panel (b) of Figure 2). This indicates that a substantial portion of legal actions are aimed at obtaining services that should be readily accessible without court intervention.¹⁴ Third, between 72.2% to 86% of tutelas receive a favorable initial verdict for

12. Unless stated otherwise, the statistics reported in this section are taken from reports produced by the Colombian Ombudsman office (Defensoría del Pueblo 2004, 2007, 2009, 2010, 2011, 2012, 2013, 2015) and the Ministry of Health and Social Protection (Minsalud 2015).

13. The extent to which individuals were using tutela writs invoking the right to health during the early 2000s led the Colombian Constitutional Court to make a major pronouncement: Ruling T-760 of 2008. In the ruling, the Court provided a thorough analysis of the problems that led to the judicialization of the health care system and ordered a set of changes, including i) full clarity about the procedures and medications that were included in the benefit packages; ii) periodical audits of the insurers in order to inform users of their performance; iii) strengthening the regulatory agency overlooking the system; and iv) a revision of the capitation payments that had been unchanged for several years. The number of tutelas invoking the right to health declined temporarily between 2008 and 2009, but by 2015 they were back to the pre-ruling levels. We can only study the period after the 2008 pronouncement because the tutela data are only available after then.

14. The remaining 35% of claims are also for medically prescribed services.

the claimant (Panel (c) of Figure 2). This suggests that the courts frequently uphold the legitimacy of these citizen appeals. Fourth, despite roughly equal enrollment proportions in each regime, only 12.7% to 36.7% of tutelas filed against insurers originate from individuals in the subsidized regime (Panel (d) of Figure 2). Thus, the low-income population is less likely to use this judicial instrument to protect their rights.

Complaints to the Supersalud. Tutela writs are a powerful legal instrument in Colombia, but they are not the only mechanism that people have to enforce access to restricted medical services. Individuals can also make complaints (and requests) to the National Health Superintendent (Supersalud), the agency in charge of overseeing the health care system. The Supersalud can apply sanctions against insurers, ranging from economic penalties to removing the license to operate, and can mediate disputes between insurers and enrollees regarding access to requested services. The complaints process is simple: citizens can file complaints to the Supersalud in person, on the telephone, in writing, through an online platform, or using the agency’s social network accounts (Facebook and Twitter). The Supersalud notifies the defendant, who is asked to respond to the complaint. If mediation is necessary, the Supersalud decides on the case within ten days.

Between 2013 and 2017, the years for which the data on complaints is available to us, people made 209,300 complaints annually against insurers, an average of 4.8 complaints per 1,000 enrollees in the system per year. This frequency is more than twice the rate of tutelas within the same time frame (Panel (b) of Figure 1). The greater incidence of complaints in comparison to tutelas could be attributed to two main factors: firstly, lodging a complaint is more straightforward, given the multiple channels provided by Supersalud, whereas filing a tutela necessitates drafting and submitting a document to a judge. Secondly, a successful tutela requires not just proof of hindered access to healthcare but also that such an impediment infringes upon the individual’s right to health. Nevertheless, tutelas and complaints are different instruments that can be used to enforce access to restricted care services. Panel (a) of Figure 3 presents a scatter plot correlating the rates of tutelas and complaints for each insurer by municipality by year.¹⁵ The estimated correlation coefficient between these two measures is 0.378.

15. For the period 2013 to 2016, where data for both tutelas and complaints are available, there are 45,403 data points at the insurer by municipality by year level.

4 Data Sources

We leverage five administrative data sources, each providing different levels of granularity and scope. These data are available either at the individual level for the entire population (vital statistics), as averages by insurer $\times$ municipality $\times$ year (enrollees, tutelas, complaints), or as averages by municipality $\times$ year (health services register). Colombia has 1,112 municipalities and, on average, 29-45 insurers operating each year by regime. Insurers operating in both the subsidized and contributory regimes are treated as separate entities. The most comprehensive dataset covers the period from 2010 to 2017.

Vital Statistics. To quantify the population-level health implications of service restrictions, we analyze mortality rates—overall, cause-specific, age-specific, and sex-specific—aggregated at the insurer by municipality by year level from 2010 to 2017. These rates are derived from individual mortality records. We use mortality because it is one of the few objective, well-measured health outcomes available over time (Finkelstein and McKnight 2008). Mortality is particularly informative for our analysis because vital statistics registers provide detailed information, including the deceased’s municipality of residence and their associated insurer, if any.

Enrollment. We use individual-level administrative data detailing the insurer with which each person was enrolled at the start of each year from 2010 to 2017. When the analysis requires, we aggregate this data to the appropriate analytical unit. This dataset originates from the registries maintained by the Ministry of Health and Social Protection.

Tutelas. We use the number of tutelas filed against health insurers and invoking the right to health. We obtained these data from the ombudsman’s office after we filed a formal petition supported by right-to-information legislation. The data covers the universe of health-related tutelas put forward to Colombian courts between 2010 and 2016. The information is available at the insurer $\times$ municipality $\times$ year level. At this disaggregated level, the data are restricted to the number of claims made, so we cannot differentiate by the type of medical care requested or whether the claims were successful.

Complaints. We use the universe of complaints to the National Health Superintendent from 2013-2017. We can identify which insurance company is involved in the complaint, the main (broad) reason for the complaint, and the age and sex of the claimant. We only use complaints against health insurers. We further exclude complaints about administrative procedures unrelated to the provision of health services and any general information requests. We also obtained these data following a right-to-information request.

Utilization of Health Services. We use data that contain all medical services effectively supplied in each municipality $\times$ year during 2010–2017, obtained from the Ministry of

Health and Social Protection of Colombia and publicly available through an online platform. This information can be disaggregated by type of service: consultation, hospitalization, procedure, or emergency. This information cannot be disaggregated by insurer.

5 Theoretical Framework

In this section, we present a simple decision model that captures the salient features of Colombia’s healthcare system. The model aims to identify the factors influencing insurers’ decisions to restrict access to services and the factors prompting enrollees to file tutelas. This framework also helps clarify our empirical strategy.

Setup. The model has two periods. In the initial period, individuals (indices omitted for simplicity) observe their health status, denoted by the discrete random variable $h_0 \geq 0$, where higher values indicate better health. If required by their health condition, they get prescribed a treatment $x \in X$ from a general practitioner, where X is the set of available treatments within the system. Each prescribed treatment must be approved by the individual’s health insurer. We assume a one-to-one correspondence between health status and prescribed treatments. The distribution of x , representing the relative demand for treatments across the population, is a known probability mass function $f(\cdot)$.

The health status in the second period, h_1 , depends on access to care:

$$h_1 = \begin{cases} h_1(x) + \eta(x) & \text{if treated} \\ h_1(0) & \text{if untreated} \end{cases}$$

where $h_1(x)$ and $h_1(0)$ are the expected health outcomes for an individual with an initial health status h_0 who has received care or not, respectively, and $\eta(x)$ represents heterogeneous effects of the treatment that are i.i.d. with mean zero and a CDF $G(\cdot)$. This term captures the uncertainty about the efficacy of medical treatment across the population which is characteristic of medical care (Gruber 2022). We assume $\eta(x)$ is known to the individual, or more precisely to their GP, but not to the insurer, who only knows its distribution, reflecting information asymmetries in the insurance market. We further assume that $h_1(x) \geq h_1(0)$, meaning that, on average, treatment is not detrimental. Finally, individuals only care about their health and wish to maximize a utility function $U(h_1)$ that is increasing in h_1 .

Insurers (indices omitted) can either authorize or restrict prescribed treatments. They receive a fixed payment y per enrollee and incur a cost $c(x)$ if they authorize a treatment x . The cost $c(x)$ varies across insurers because each insurer negotiates contracts independently

with its network of providers. If an insurer denies or delays authorization, the individual may file a tutela to enforce access to the treatment. Although filing a tutela is free of charge, individuals bear a cost δ in health units, which accounts for the time and stress involved in litigation, or the effort required to obtain the necessary information to file a tutela correctly. Insurers also face an insurer-specific administrative cost γ of handling a tutela, irrespective of its outcome. Lastly, there is an exogenous probability $\beta(x)$ that the court ruling will be in favor of the plaintiff.

Optimal Choices. Comparing the expected utility with and without litigation, an individual facing restrictions on access to x will file a tutela if

$$\beta(x)U(h_1(x) + \eta(x) - \delta) + (1 - \beta(x))U(h_1(0) - \delta) \geq U(h_1(0)), \quad (5.1)$$

or, equivalently, if

$$\beta(x) \geq \frac{U(h_1(0)) - U(h_1(0) - \delta)}{U(h_1(x) + \eta(x) - \delta) - U(h_1(0) - \delta)}. \quad (5.2)$$

The right-hand side of this inequality is decreasing in the idiosyncratic health gain from the treatment, $\eta(x)$. Thus, if $\beta(x) > 0$, there exists a threshold $\underline{\eta}_x$ such that individuals with $\eta(x) \geq \underline{\eta}_x$ will litigate when facing barriers. This threshold is treatment-specific and depends on the average health gains, $h_1(x) - h_1(0)$: higher expected health gains require a lower threshold to motivate individuals to litigate. Given the distributional assumptions, the fraction of individuals that would file tutelas for x is

$$\alpha_x = (1 - G(\underline{\eta}_x)) \cdot f(x). \quad (5.3)$$

Insurers can estimate the probability of litigation for each service and use this information to decide which treatments to restrict. Specifically, based on a cost-benefit analysis, it is optimal to impose barriers to a treatment x if

$$\alpha_x \left(\beta(x)(c(x) + \gamma) + (1 - \beta(x))\gamma \right) \leq c(x), \quad (5.4)$$

or, equivalently, if

$$c(x) \geq \frac{\alpha_x}{1 - \alpha_x \beta(x)} \gamma, \quad (5.5)$$

meaning that the costs are sufficiently large after considering the probability of litigation and a favorable ruling. Any treatment x that satisfies this inequality will be restricted, regardless of the requester.

Figure 4 presents the results of a simulation of the optimal choices made by individuals and insurers, as derived from Equations (5.2) and (5.5).¹⁶ The figure illustrates how the cost threshold—beyond which a treatment is denied, indicated by the red line—changes across various simulated treatments that differ in their expected health benefits, while keeping all other parameters constant.

There are three main takeaways. First, services that are more likely to face restrictions are those where the expected health benefits are low relative to their costs, as individuals are less inclined to pursue litigation in such cases. This implies that, from a cost-benefit perspective, certain restrictions might be seen as optimal. The need to curb healthcare spending on services that yield low benefits has been a significant topic of debate, especially given that healthcare spending continues to rise in most countries, regardless of how their healthcare systems are financed and organized (Chernew and Newhouse 2011; OECD 2015). In settings where insurance is provided by Health Maintenance Organizations (HMOs) that can adjust plan offerings, risk premiums, or set fee-for-service rates, financial incentives exist for users to rationalize their use of services or internalize the associated costs. However, in Colombia, the mandatory health plan is uniform and comprehensive, capitation rates are fixed, and co-payments apply only to a small segment of the highest-income population. As a result, the rationalization of service use is often achieved through barriers to access.

The model also shows that some services with significant health benefits may also face restrictions. The optimality of these restrictions is not straightforward, as they can also impose a heavy financial burden on the system. However, in Colombia, the comprehensive right to health generally takes precedence. The Constitutional Court has consistently prioritized access to healthcare over financial considerations, emphasizing the protection of individuals' health rights (Yamin and Parra-Vera 2009; Arrieta-Gómez 2018). This creates a tension between upholding individual health rights and ensuring the financial sustainability of the healthcare system.

We cannot assess the optimality of restrictions with the available data, as we lack detailed qualitative information on the types of services requested in claims at a disaggregated

16. For details of the simulation, see Appendix A.1.

level. However, analysis from the Ombudsman’s office, based on a sample of tutelas (see Figure A.3), indicates that restrictions occur across all types of services and for various diagnoses. This suggests that both low- and high-benefit care can face access barriers, which is consistent with the model’s predictions. In Section 9, we use diagnoses classified under ICD-10 codes from vital statistics data to examine whether restrictions are associated with worse comorbidities among individuals at the time of death. If more severe comorbidities indeed reduce the effectiveness of care, and insurers are primarily focused on rationing ineffective care, we would expect a higher prevalence of restrictions in this population.

A second takeaway is that, although restrictions impact all individuals prescribed a restricted service x , those who choose to litigate are likely to be the ones who stand to benefit the most (i.e., those with higher η_x values). Given that most claims result in a favorable outcome for the plaintiff, litigation may serve as an effective mechanism to secure access to essential care.

Finally, even under similar healthcare demands, restriction rates will vary across insurers—and within the same insurer in different geographical locations—due to differences in their cost structures and management practices. This variation is central for our identification strategy. For instance, an insurer-specific cost shock in one location could affect the prevalence of supply-side barriers in another location where the same insurer operates. This creates supply-driven variation in restriction rates that is spatially correlated.

Restrictions and Tutela Rates. We can now characterize some of the aggregate quantities used in the empirical analysis. The model predicts that some services will invariably encounter supply-side barriers to access by an insurer if the costs are large enough. Let us denote the subset of services restricted by $X^r \subseteq X$. The proportion of enrollees encountering restrictions equates to the potential proportion of enrollees prescribed those services:

$$Restriction = \sum_{x \in X^r} f(x). \quad (5.6)$$

Consequently, the fraction of enrollees that will file a tutela is given by

$$Tutela = \sum_{x \in X^r} \alpha_x. \quad (5.7)$$

where α_x is given by (5.3) and $\alpha \equiv \frac{\sum_{x \in X^r} \alpha_x}{\sum_{x \in X^r} f(x)}$ is the fraction of enrollees filing tutelas conditional on their requested services being restricted. It follows that

$$Tutela = \alpha Restriction. \quad (5.8)$$

This equation says we can approximate the unobserved restriction rate using the tutela rate. This is because the latter is always a fraction of the former. Moreover, as restrictions will, on average, deteriorate population health, a supply-side-driven rise in the restriction rate will be accompanied by an increase in the tutela rate and lower population health. The effect will be attenuated by the fact that a fraction

$$ClaimsUpheld = \sum_{x \in X^r} \beta(x) \alpha_x \quad (5.9)$$

of services are delivered after favorable judicial rulings. However, the rise of restrictions signals an increase in unmet healthcare demand, defined as the portion of requested but undelivered services. Specifically,

$$\begin{aligned} unmetDemand &= Restriction - ClaimsUpheld \\ &= (\alpha^{-1} - \beta) Tutela, \end{aligned} \quad (5.10)$$

where $\beta = \frac{\sum_{x \in X^r} \beta_x \alpha_x}{\sum_{x \in X^r} \alpha_x} \in [0, 1]$ and $(\alpha^{-1} - \beta) \geq 0$. In sum, tutela rates track unmet demand.

Panel (b) of Figure 3 shows the raw correlation between the (log) tutela and (log) mortality rates using municipality $\times$ insurer $\times$ year data. Panel (c) replicates the exercise using the complaint rate. Consistent with the model predictions, there is a clear positive association. We argue that this reflects the impact of restricting access to health services on health outcomes, as described in Equation 5.8, which results in an unrealized demand for health care, as described in Equation 5.10. We now proceed to discuss the empirical strategy used to test this hypothesis.

6 Empirical Strategy

We focus here on tutela claims for expositional purposes, but each model is also estimated using complaints to the National Health Superintendent. Each observation corresponds to an average over the enrollees in the insurer-municipality-year triplet, multiplied by 1,000. In our estimations, we weight observations according to the enrollee count within these triplets. However, results are qualitatively unchanged if we omit weights.

6.1 Supply Restrictions and Mortality: Three-Way Fixed Effects and IV

We first estimate a model of the form:

$$Mort_{i,m,t} = \tau Tutela_{i,m,t} + \delta X_{i,m,t} + \phi_{m,t} + \theta_{i,t} + \varphi_{i,m} + \epsilon_{i,m,t}, \quad (6.1)$$

where $i \in \{1, \dots, I\}$ indexes health insurers, $m \in \{1, \dots, M\}$ indexes municipalities, and $t \in \{1, \dots, T\}$ indexes years. $Mort_{i,m,t}$ is the mortality rate, $Tutela_{i,m,t}$ is the rate of tutela claims, $\phi_{m,t}$ are municipality-year fixed effects, $\theta_{i,t}$ are health insurer-year fixed effects, and $\varphi_{i,m}$ are municipality-insurer fixed effects. $X_{i,m,t}$ are time-varying covariates, discussed below.

The main identification challenge is isolating the tutela rate variation induced by supply-side restrictions from demand-driven changes. If that is achieved, the parameter τ can be interpreted as the effect of supply-side restrictions to medical care access on mortality. In the theoretical framework, we derived that $Tutela = \alpha \text{Restriction}$ (see Equation (5.8)), where $\alpha \in [0, 1]$ represents the fraction of enrollees who file tutelas (i.e., litigate) when faced with restrictions. This equation highlights the need to consider factors influencing both the restriction rate and the prevalence of litigation.

The fixed effects account for most potential demand-side drivers. For example, the municipality-by-year fixed effect, $\phi_{m,t}$, captures local health conditions, the availability and quality of healthcare providers, demographic changes, and geographic epidemiological shocks (i.e., the location-specific distribution of $f(x)$ in the model). This fixed effect also accounts for local cultures of claiming, knowledge about enforcing the right to health, and the leniency of local courts (i.e., the location-specific distribution of δ and $\beta(x)$ in the model). The insurer-by-municipality fixed effect, $\varphi_{i,m}$, captures time-invariant cross-sectional heterogeneity in risk profiles and the propensity to litigate. This does not address dynamic selection related to changes in insurers or municipalities of residence, but we show that there is no evidence of this in our setting. Lastly, $\theta_{i,t}$ controls for insurer-level shocks that impact service quality and management practices, including system-wide changes that may affect insurers differently.

We address the concern that there remain relevant omitted variables that vary at the insurer-municipality-time level by additionally introducing an instrumental variables approach. To construct an instrumental variable, we leverage the fact that insurers operate across multiple municipalities. The key idea is that there are insurer-specific (supply-side) factors that determine the restriction rate and that are orthogonal to *local* (municipality-level) demand for health care. Natural examples of insurer-specific supply-side factors are cost structures determined by different service provider contracts and managerial practices. In terms of the model parameters, the cost structure would be captured by the distribution

of $c(x)$, and efficiency would be reflected in the value of γ .

The instrument is an insurer-specific tutela rate, calculated using data exclusively from municipalities in regions outside the department¹⁷ where municipality m is located, but where the insurer operates:

$$Z_{i,m,t} = \sum_{k:D_k \neq D_m} s_{i,k,t} Tutela_{i,k,t}, \quad (6.2)$$

where $s_{i,k,t} \equiv \frac{N_{i,k,t}}{\sum_{k:D_k \neq D_m} N_{i,k,t}}$, with $N_{i,k,t}$ representing the total number of enrollees in health insurer i in municipality k and year t . The variables D_m and D_k denote the departments where municipalities m and k are located, respectively. The exclusion restriction holds if the tutela rate in municipalities outside the department where m is located does not directly influence the mortality rate in municipality m , conditional on the structure of fixed effects described above.

In some specifications, we include two time-varying covariates $X_{i,m,t}$: the one-period lagged mortality rate, which reflects the average health outcomes of the previous year and accounts for potential mean reversion, and the one-period lagged market share of the insurer in the municipality¹⁸, as this may influence the insurer's ability to negotiate contracts with providers. Importantly, we provide direct evidence that individuals do not respond to increased restrictions by switching insurers, indicating that restrictions do not drive changes in market shares.

6.2 Supply Restrictions and Utilization of Health Services: Shift-Share - IV

The mechanism through which supply-side barriers impact health outcomes is by reducing the utilization of healthcare services. If demand-side shocks were the primary driver, we would expect an increase in healthcare utilization rather than a decline. This distinction is a key prediction in our framework. We test this prediction using a similar instrumental variable (IV) strategy. Since the health services data are only available at the municipality-by-year level, we adapt our empirical strategy for the analysis of mechanisms by employing a shift-share design. The underlying premise remains that insurer-specific restriction rates are correlated across municipalities due to supply-side factors that are unrelated to local healthcare demand. The model is specified as:

17. Departments in Colombia are the primary administrative divisions, similar to states or provinces in other countries. Colombia is divided into 32 departments, which vary widely in population, economic activity, and geographic characteristics.

18. Market share is defined as the proportion of enrollees a health insurer has within the municipality.

$$y_{m,t} = \pi Tutela_{m,t} + \gamma W_{m,t} + \phi_m + \theta_t + \epsilon_{m,t}, \quad (6.3)$$

where $y_{m,t}$ is the rate of health services provided in the municipality, which we divide into patient consultations, hospitalizations, procedures, and emergencies, all defined on a per enrollee basis. $W_{m,t}$ is the same vector of controls but at the municipality level, and ϕ_m is a municipality fixed effect. To construct the instrument, note that the municipal tutela rate is a weighted average of the tutela rates of each insurer operating in the municipality, where the weights correspond to market shares:

$$Tutela_{m,t} = \sum_i s_{i,m,t} Tutela_{i,m,t}, \quad (6.4)$$

where $s_{i,m,t}$ is the municipal share of enrollees of insurer i at time t . We can decompose $Tutela_{i,m,t}$ into an aggregate, insurer-level rate, and an idiosyncratic, municipal-level rate:

$$Tutela_{i,m,t} = Tutela_{i,t} + \mu_{i,m,t}, \quad (6.5)$$

where $Tutela_{i,t}$ is the tutela rate of insurer i at time t at the national level. In practice, we calculate this rate using data exclusively from municipalities in departments outside the department where municipality m is located. The instrument is then constructed to utilize variation from the aggregate rates ($Tutela_{i,t}$), and not from the local idiosyncratic rates ($\mu_{i,m,t}$):

$$B_{m,t} = \sum_i s_{i,m,t_0} Tutela_{i,t}, \quad (6.6)$$

where s_{i,m,t_0} is the municipal share of enrollees of insurer i in a baseline year $t_0 = 2010$. Defining the market share in the baseline year ameliorates concerns that people might switch insurers because of restrictions, which would endogenously change the risk-pool distribution.

Like all shift-share IVs, the instrument has two components: First, how exposed a municipality is to restrictions by an insurer, given by the respective market share s_{i,m,t_0} . Second, how many restrictions were imposed by each insurer on average outside the municipality m , proxied by the tutela rate. The recent literature on shift-share IV shows that the validity

of the instrument in this set-up can be argued in terms of the exposure variables (s_{i,m,t_0}) (Goldsmith-Pinkham et al. 2020), or the aggregate shocks ($Tutela_{i,t}$) (Borusyak et al. 2022). Following the discussion in the previous section, we argue that the restriction rate in municipalities outside the department where m is located is unrelated to local demand for health care, which implies exogeneity of the aggregate shocks.

7 Results: Restrictions Cramp Utilization of Medical Services

We start our analysis by examining the impact of restrictions on healthcare service utilization. As previously noted, if the variation in tutela rates is driven by demand-side factors, an increase in tutelas would lead to higher healthcare utilization. In contrast, if supply-side factors are the primary drivers, an increase in tutelas would result in a decline in utilization. This is a clear prediction that underpins our analysis and can be directly tested using the data.

Estimates of Equation 6.3 are presented in Tables 1 and 2. The first table focuses on consultations and procedures, while the second addresses hospitalizations and emergencies, all evaluated on a per enrollee basis. We present results using the fixed-effects OLS specification and the shift-share IV for both tutela and complaint rates. The Kleibergen-Paap F-statistic on the excluded instrument in the first stage exceeds 16. Descriptive statistics (mean and standard deviation) reported in the tables indicate that, on average, in a year, enrollees have 2.3 (SD 1.1) consultations, 1.77 (SD 1.12) procedures, 0.042 (SD 0.023) hospitalizations, and 0.13 (SD 0.10) emergencies.

The evidence is clear that supply-side restrictions on access to medical care reduce all types of service utilization.¹⁹ For instance, in our preferred IV specification, an increase of one tutela per 1,000 enrollees—equivalent to nearly a 50 percent rise in the tutela rate—is associated with a 13.6 percent decline in consultations, a 35.7 percent decline in procedures, a 10.9 percent decline in hospitalizations, and a 36.4 percent decline in emergency services, all relative to their respective means. These results also indicate that restrictions on access are not temporary, or not only around delay, but that there is a lot of outright denial.

19. If, for example, restrictions were predominantly on consultations, then one might imagine that as (preventive) consultations fall, there are knock-on effects that lead to an increase in hospitalizations and emergencies. Thus, the direct effects seem to dominate any knock-on effects of denials. It remains possible that there is some substitution within each of the broad categories of consultations, procedures, hospitalizations, and emergency care.

8 Results: Restrictions Impact Mortality

The estimates are presented in Table 3. We show results with and without the instrument, and with and without the time-varying controls. To improve comparison with the results on the utilization of medical services, the Table also shows the results of estimating Equation 6.3 using the municipal-year (rather than the municipal-insurer-year) mortality rate as the dependent variable. We consistently provide estimates for both tutela and complaint rates as these provide alternative sources of variation in the same underlying variable (restrictions). The Kleibergen-Paap F-statistic on the excluded instrument in the first stage is shown and, in all cases, is above 16.

An increase in supply-side restrictions on access to health services leads to a statistically significant and meaningful rise in the mortality rate, a finding that remains robust across alternative specifications and measures. Using our preferred IV specification, we estimate that an additional tutela per 1,000 enrollees is associated with an increase of between 7.6 and 11.6 percent in the mortality rate relative to the mean. Estimates using complaints follow the same direction but can be of a smaller in magnitude for the IV specification (a rise between 4.1 and 14.7 percent), possibly reflecting that tutelas are typically used for more critical conditions, as they require the plaintiff to show that their right to health is being breached.

Although not directly comparable²⁰, to provide some benchmarking for these effect sizes, we note that studies analyzing Medicaid expansion in the US have found impacts of health insurance coverage on mortality in a wide range, from no effect (Finkelstein and McKnight 2008; Weathers and Stegman 2012) to a reduction in all-cause mortality of between 6.1 percent (Sommers et al. 2012) and 9.4 percent (Miller et al. 2021). Goodman-Bacon (2018) estimates that, after the introduction of Medicaid, mortality among nonwhite children on Medicaid fell by 20 percent, leading to a reduction in aggregate nonwhite child mortality rates of 11 percent. More recently, Goldin et al. (2020) exploits experimental variation from a randomized outreach study finding that the average per-month effect of getting health insurance on two-year mortality was approximately -0.18 percentage points among middle-aged adults.

In summary, results show that restrictions significantly affect average health outcomes, even on extreme outcomes like the mortality rate. The actual impact of restrictions on population health is likely greater, as it includes unmeasured effects on morbidities that do not directly translate into mortality.

20. The magnitudes are not comparable as the expansions or extensions of health insurance coverage in these studies are of different magnitude than the contraction of effective coverage that is captured by an increase in the tutela rate.

Dynamic Effects. Table 4 shows estimates of the baseline three-way fixed effects specification shown in Equation 6.1, but using past (one-year lagged) tutela or complaint rates. These would affect health outcomes if the impact of restrictions takes time to kick in, so results from this exercise are informative of the timing of effects. We find no significant impact of lagged rates, which shows that barriers to access have immediate (within the year) impacts on mortality.

Placebo. Table 4 also shows estimates with future supply-side restrictions, captured by the one-year forward rates, on the right hand side. These should not affect current health outcomes, so we expect a null result in this regression. This is what we find. If forward rates had turned out to be significant, this would have flagged that we have failed to measure a relevant unobservable trend at the insurer $\times$ municipality $\times$ year level.

Dynamic Selection. If individuals switch insurers due to restrictions, the risk profile of enrollees may change over time, posing a threat to our identification strategy. For instance, if those who switch are in worse health—a likely scenario when faced with increased supply-side barriers—the remaining enrollees will be healthier. This would render our estimates of a positive correlation between mortality rates and restrictions conservative. Conversely, if healthier individuals are the ones switching, the remaining enrollees would be in worse health, potentially making the positive correlation between mortality rates and restrictions spurious.

We analyze dynamic selection by examining whether the probability of an individual switching insurers is correlated with the restriction rate. To conduct this analysis, we obtained individual-level administrative data detailing each individual’s insurer enrollment at the beginning of the year. We then estimate a model where the dependent variable is an indicator equal to one if an individual changed insurers between year $t - 1$ and t , and the independent variable of interest is the tutela or complaint rate in year $t - 1$. We control for individual fixed effects and flexible trends for initial insurer $\times$ year and initial municipality $\times$ year, where “initial” refers to the insurer or municipality of the individual at her first appearance in the data.

The estimates using the full sample are reported in Table 5, while Table 6 repeats this analysis, dividing the sample by sex and age groups. On average, 14 percent of all individuals change insurers in a year. However, we find no evidence that individuals switch insurers in response to increased restrictions. This result holds regardless of the individual’s age, sex, or the subsystem (contributory or subsidized) in which the insurer operates. These findings suggest that dynamic selection is unlikely to drive our results.

Finally, Appendix Table A.2 presents results from a similar analysis examining whether an individual’s probability of changing their municipality of residence is correlated with the restriction rate. On average, between 4 percent (SR) and 7 percent (CR) of individuals change their municipality of residence in any given year. However, the likelihood of migrating is not associated with increases in restriction rates in the year prior to the move. In one specification, we find a marginally statistically significant effect, but its magnitude is small: a one standard deviation increase in the tutela rate leads to less than a one-tenth of a percentage point increase in the migration rate, which is already low on average. These results make it unlikely that dynamic selection from geographic mobility is driving our results.

8.1 Heterogeneity in Impact

By Regime. In Table 7 we repeat the analysis, dividing the sample between the contributory (CR) and subsidized (SR) regimes, which have different target populations, the SR covering the most economically vulnerable segments of the population. Earlier we noted (Panel (b) of Figure 1) that tutela and complaint rates are significantly larger in the CR than the SR, which suggests that the CR population either faces more restrictions or has a greater tendency to use judicial instruments. It is hard to identify which, but both are plausible. We show only the specifications with the full set of controls. In both regimes, we find evidence that supply-side restrictions lead to an increase in mortality rates. The IV results indicate that, in the CR, an additional tutela or complaint per 1,000 enrollees leads to increases of between 6.0 and 10.4 in the mortality rate. In the SR, a similar change in each of the two measures leads to an increase of 13.1 and 16.0 percent in the mortality rate. However, the complaint rate estimates for the SR are imprecise.

By Cause of Death. A notable feature of judicial claims is that they pertain to all forms of medical care, supported by a wide range of medical diagnoses. Panel (a) of Figure A.3 shows that of all tutelas between 2000 and 2015, 9.3% requested appointments with health specialists, 12.8% requested exams, 15.5% surgeries, 17.1% medications, and 18.0% treatments. Moreover, a recent report by the Ministry of Health and Social Protection (Minsalud 2015) shows that 11.6% of tutelas are supported by medical diagnoses of neoplasia, 9.9% refer to diseases of the nervous system, and 8.7% to diseases of the circulatory system (Panel (b) of Figure A.3).

Although denials appear pervasive, some health conditions require more urgent or comprehensive care, making it plausible that the impact of restrictions varies by cause. We therefore investigate cause-specific mortality rates; see Figure 5. The data on tutelas and complaints by cause are only available aggregated at the national level, so we use the

municipality-insurer-year level rate as before. To facilitate exposition, we focus on the 15 most prevalent causes of death according to the classification of diseases defined by the Colombian National Statistical Agency, which has 71 categories, and we only report results for the IV specification using tutela rates. The Figure displays both the coefficient estimates and the percent change relative to the average cause-specific mortality rate.

There are two main takeaways from this exercise. First, most of the point estimates are positive, and many are statistically significant at standard levels, indicating that restrictions generally increase mortality rates across various conditions. This aligns with the wide range of diagnoses supporting the tutela claims. Second, some conditions exhibit a stronger response to restrictions. For instance, mortality from stomach and colon cancer is estimated to increase by 7.7 and 9.1 percent, respectively. These two conditions are known to be particularly deadly if not diagnosed and treated early.

By Age and Sex The information on complaints is available at the individual level, allowing us to identify the age group and the sex of the person affected in the complaint (who is not necessarily the person who files the claim), so we can create group-specific rates. The information on vital statistics is also available at the individual level so that we can construct group-specific mortality rates. We estimate the three-way fixed effects model for different subsamples by combining the two sources.²¹

As a starting point, Panels (a) and (b) of Figure A.4 show the average complaint and mortality rates for males and females and four age groups: 0-5, 6-24, 25-49, and 50 or above. Two patterns are evident in the data: first, complaint rates have a U-shaped pattern across age groups. The lowest rate is at ages 6-24, while the largest is at age 50-plus. Second, the complaint rate tends to be larger for females. In both cases, differences could be explained by a higher propensity to make complaints or by a higher prevalence of restrictions.

The estimated effects are presented in Figure 6. For clarity, we report results only for the instrumental variable specification with the full set of controls. The impact on mortality rates is estimated at 9.0% for individuals aged 6 to 24 and 2.4% for those over 50, both measured relative to the average and based on an increase of one additional complaint per 1,000 enrollees. The effect of restrictions appears slightly larger for males than for females, though the difference is not statistically significant at conventional levels.

By Competition in the Insurance Market A mechanism that could explain the prevalence of supply-side restrictions is limited competition between insurers within municipal-

21. We cannot identify age and sex in the tutela data, but the impacts of tutela and complaint rates on aggregated mortality rates were similar.

ities. As shown in Panel (a) of Appendix Figure A.2, the average number of insurers per municipality was 6.5 in the CR, and 4.2 in the SR, but these numbers conceal important heterogeneities: approximately 10 percent of municipalities have a single insurer in the SR (Panel (b) Appendix Figure A.2), while between 10 and 20 percent of all municipalities have at most two insurers in the CR (see Panel (c) of Appendix Figure A.2). Moreover, the average cross-municipality Herfindahl-Hirschman Index (HHI) of market concentration, defined in terms of the share of enrollees, is 5,251 in the contributory regime and 5,463 in the subsidized regime. This is more than twice the lower threshold used by the Justice Department of the United States to classify an industry as highly concentrated.

To investigate the role of competition, we estimated Equation 6.3 interacting the tutela or complaint rates with the HHI index, or with alternative binary classifications of municipalities based on the HHI index. We find no clear evidence that impacts of restrictions on mortality emerge predominantly from low-competition markets- the interaction term is, in general, statistically insignificant at standard levels across specifications.²² This is not entirely surprising. For example, higher market concentration can generate economies of scale or increase insurers' power to negotiate lower prices with providers, reducing the incentive to use denials to control costs. These trade-offs have been studied in the literature on the industrial organization of health care markets (Gaynor et al. 2015; Ho and Lee 2017), and they suggest that enforcing more competition may not be enough to induce a better provision of services.

Overall, the results described in this section provide compelling evidence that restrictions on access to health care impose significant health costs on the population. The identified increases in mortality rates are robust to varying the controls (two and three-way fixed effects models, time-varying controls), the estimator (with and without IV, using the two-way or three-way variation), and the measure of restrictions (tutela vs. complaint rates). This stability of the estimates mitigates concerns that selection drives our results, and we find no evidence of endogenous mobility across insurers. In the next section, we investigate the impacts of restrictions on the likely mechanism- the utilization of health care services. This allows us not only to verify the mechanism in principle but also to identify the size of these effects, and their distribution by service type.

22. We also find that the descriptive association of tutelas with the HHI is not statistically significant. Results are available on request.

9 Comorbidities and Mortality Rates

The theoretical framework presented in Section 5 suggests that, from a cost-benefit perspective, some restrictions on access to medical care can be warranted. Due to the financial constraints of health systems, it is not feasible to provide every requested service, making prioritization essential. This is particularly relevant in cases where providing certain services is unlikely to alter the mortality outcomes for patients.

We cannot directly determine whether an individual who faced supply-side barriers to care would have survived if they had received medical attention. However, mortality records provide data on the comorbidities present at the time of death. Since certain comorbidities are strong predictors of mortality, we can examine whether tutela and complaint rates are associated with the prevalence of these comorbidities. A positive correlation would suggest that barriers are more likely to target individuals with a high probability of mortality. Conversely, if tutela and complaint rates are also associated with comorbidities that are not strong predictors of mortality, this would imply that restrictions are applied more broadly, without specific targeting. It is important to note that this would only be suggestive evidence, as the presence of restrictions could also exacerbate comorbidities.

For the analysis, we use the Age-Adjusted Charlson Comorbidity Index (CCI), a widely adopted tool in medical practice designed to predict ten-year mortality for patients with various comorbid conditions. The CCI covers 17 conditions, each weighted based on its association with mortality risk. These conditions include heart disease, AIDS, liver disease, and various types of cancer, with each assigned a score corresponding to its relative risk of death. The Age-Adjusted CCI further refines this score by adding points based on the patient’s age, assigning one additional point for each decade over 40 or 50 years.

We begin by calculating the CCI for each deceased individual based on all reported comorbidities at the time of death, which can include up to twelve ICD-10 codes. We then divide the sample into five groups according to their CCI scores. The first group includes individuals with CCI scores of two or lower, indicating few or no significant comorbidities. The final group comprises individuals with CCI scores above nine, reflecting multiple or severe conditions that substantially increase mortality risk. For each group, we calculate a CCI-specific mortality rate, defined as the number of deaths per 1,000 enrollees. Using these newly defined mortality rates, we then estimate Equation 6.1.

The results in Table 8 reveal a consistent pattern: higher tutela and complaint rates are associated with increased mortality rates across all CCI groups. For individuals with lower CCI scores (≤ 2), a unit increase in the tutela rate corresponds to an 8.16 percent increase in mortality rates in our preferred IV specification. For those with higher CCI scores (≥ 9),

the effect is slightly smaller, at 6.1 percent.

Overall, the findings indicate that restrictions are not strongly targeted based on individuals' mortality risk; instead, there is a general increase in mortality rates across all groups. This aligns with our theoretical framework, which posits that medical care with both low and high expected benefits is likely to face restrictions. However, the results also suggest that reducing barriers could potentially alter the mortality outcomes for some patients.

10 Conclusion

Our results show that supply-side restrictions on access to health care services manifest as substantial reductions in medical consultations, hospitalizations, emergency care, and medical procedures. In line with this, we identify increases in mortality rates. The estimates are robust to varying the estimator and the measure of restrictions (tutela vs. complaint rates). There is no evidence that citizens migrate across insurers in response to restrictions, mitigating concerns over dynamic selection. The estimates are robust to allowing for mean reversion, and for other relevant time-varying covariates. They are pervasive across social groups defined by economic and demographic characteristics, and across causes of death. They do not vary by insurer-concentration indices.

Our results are relevant to emerging health care insurance systems in many developing countries. Identifying an effective design for these new regimes is important because public budgets are tight, and health and economic status are more intricately tied among the poor. Our results are also more widely relevant as richer countries increasingly have to grapple with rising demand that current provisions are, in general, unable to satisfy.

Colombia has made a constitutional commitment to the right to health and acted to implement universal health care coverage ahead of many other countries. It has in place a well-functioning, widely accessed institution designed to protect citizen rights. However, the organization of the delivery of health care services, which operates through insurance companies (mostly but not entirely privately owned), has design flaws. There is too much regulation insofar as insurers are not allowed to set prices or select enrollees and, at the same time, too little regulation insofar as insurers are not penalized when complaints against them are received and upheld. Our estimates provide a lower bound on the population-level health gains that can be achieved through re-designing the system.

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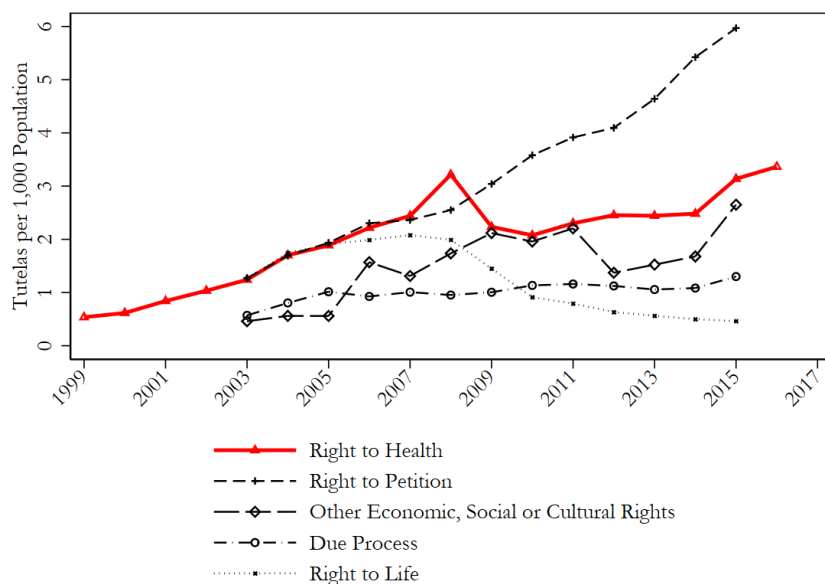
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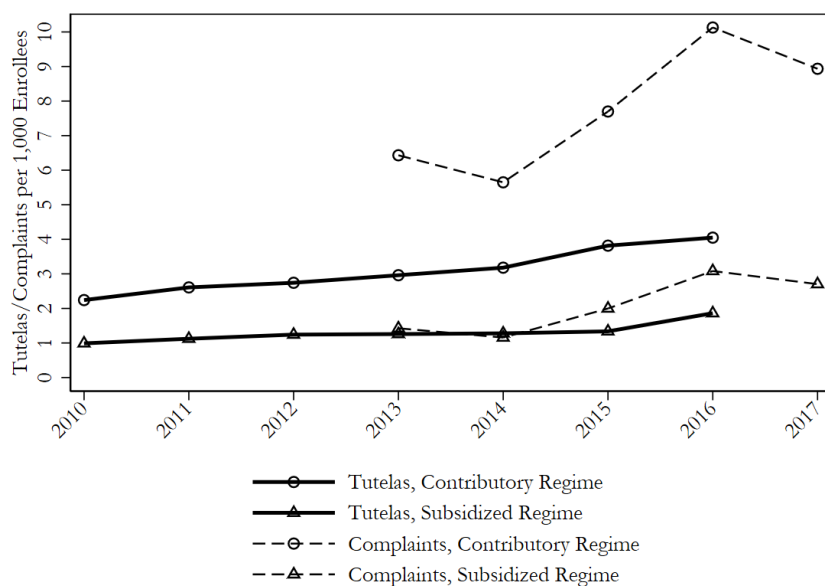
Tables and Figures

Figure 1: Trends in Tutelas and Complaints

(a) Tutelas per 1,000 Inhabitants by Fundamental Right Invoked



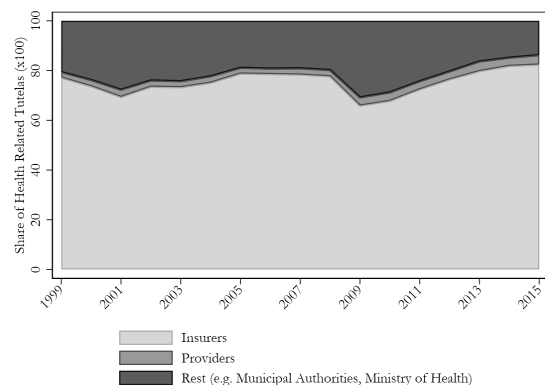
(b) Complaints and Tutelas Against Health Insurers per 1,000 Enrollees by Regime



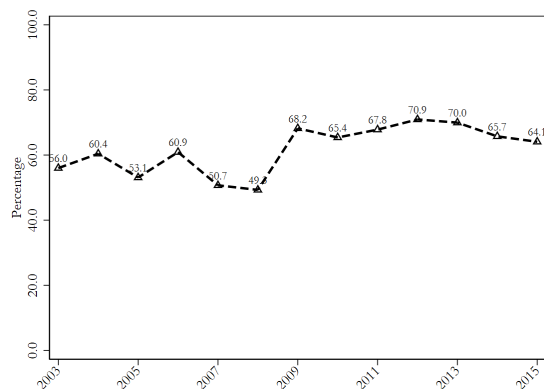
Notes: The numbers reported in the right-to-health series of Panel (a) include complaints made against other actors in the system different from insurers (i.e., IPSs, Secretary of Health and Social Protection, local authorities). See footnote 13 for a discussion on the peak of health related tutelas in 2008. The numbers in Panel (b) are based on Defensoría del Pueblo (2004, 2007, 2009, 2010, 2011, 2012, 2013, 2015), and include only complaints and tutelas made against insurers.

Figure 2: Evolution of Tutelas Invoking the Right-to-Health by Defendant Body, Share of Services Requested Included in the Benefit Package, First Instance Decision by the Judge, and Regime

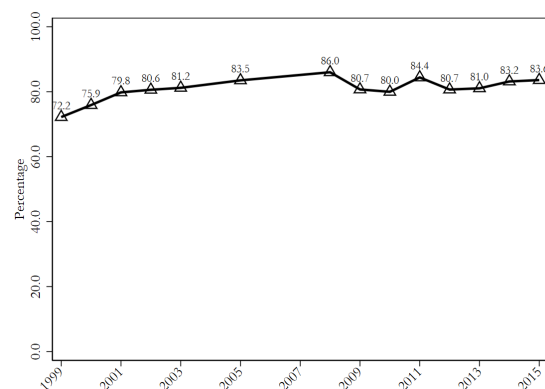
(a) Share of Health Related Tutelas by Defendant Body



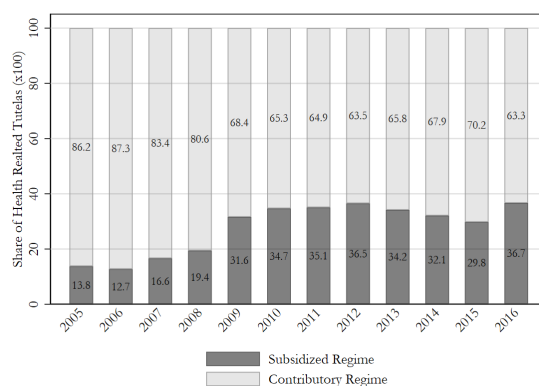
(b) Share of Services Demanded in Tutelas that are Included in the Benefit Package



(c) Share of Health Related Tutelas Decided in Favor of the Claimant

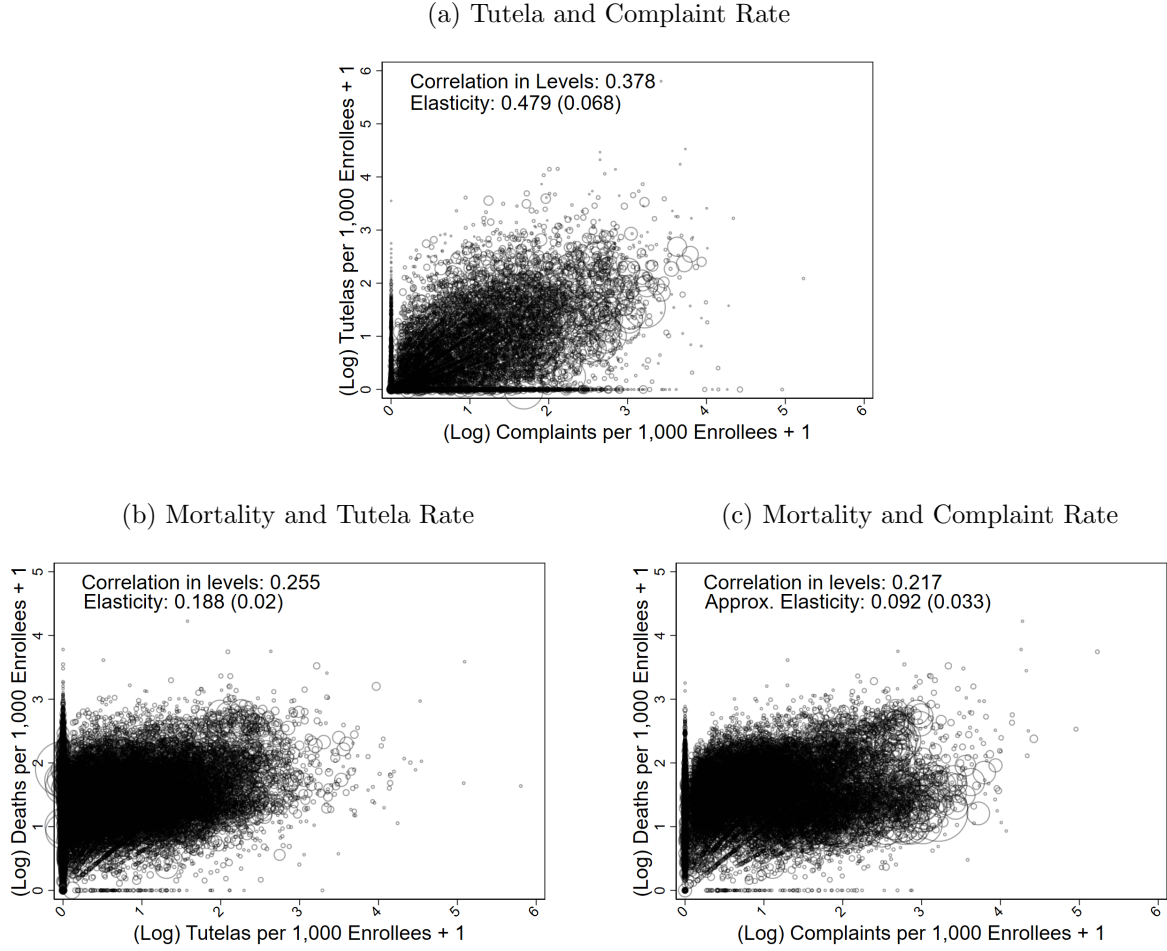


(d) Share of Health Related Tutelas by Regime



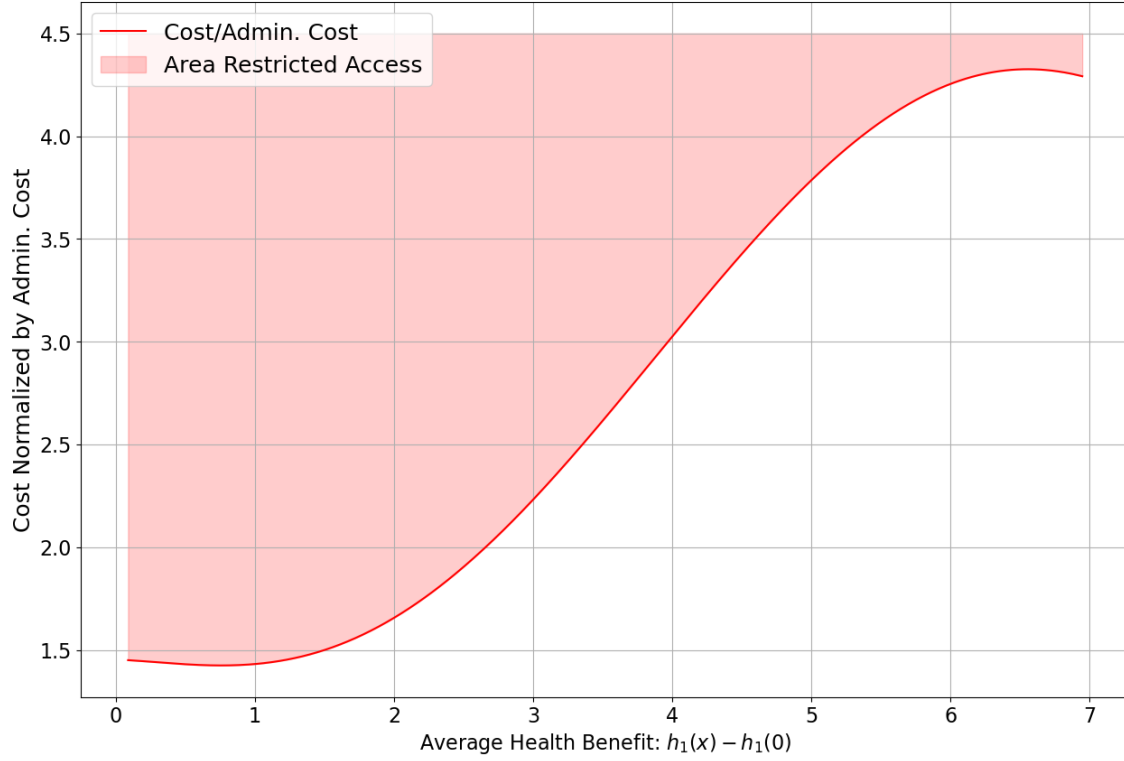
Source: Authors calculation based on Defensoría del Pueblo (2004, 2007, 2009, 2010, 2011, 2012, 2013, 2015). The numbers reported in Panels (c) and (d) include tutelas made against other actors in the system different from insurers (i.e. IPSSs, Secretary of Health and Social Protection, local authorities).

Figure 3: Correlations between Mortality, Tutela and Complaint Rates



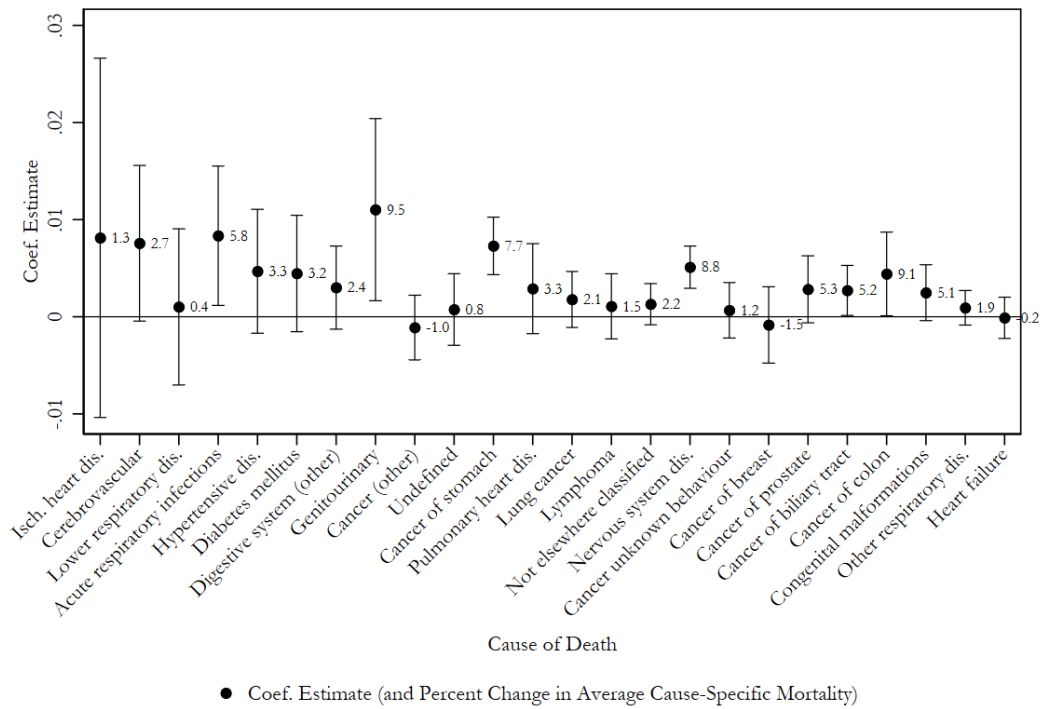
Notes: The scatter plot in Panel (a) shows the relation between the (log) of the tutela and (log) complaint rates (plus one) defined at the municipality $\times$ insurer $\times$ year level. The scatter plot in Panel (b) shows the relation between the (log) of the tutela and (log) mortality rates (plus one) defined at the same level. Panel (c) repeats the exercise but using the complaint rate. The size of each circle is proportional to the total share of enrollees. The approximate elasticity is measured by the estimated slope of a regression of the respective variables. The sources of data to construct the figure are described in Section 4.

Figure 4: Expected Net Benefits, Service Cost, and Decision to Restrict a Service



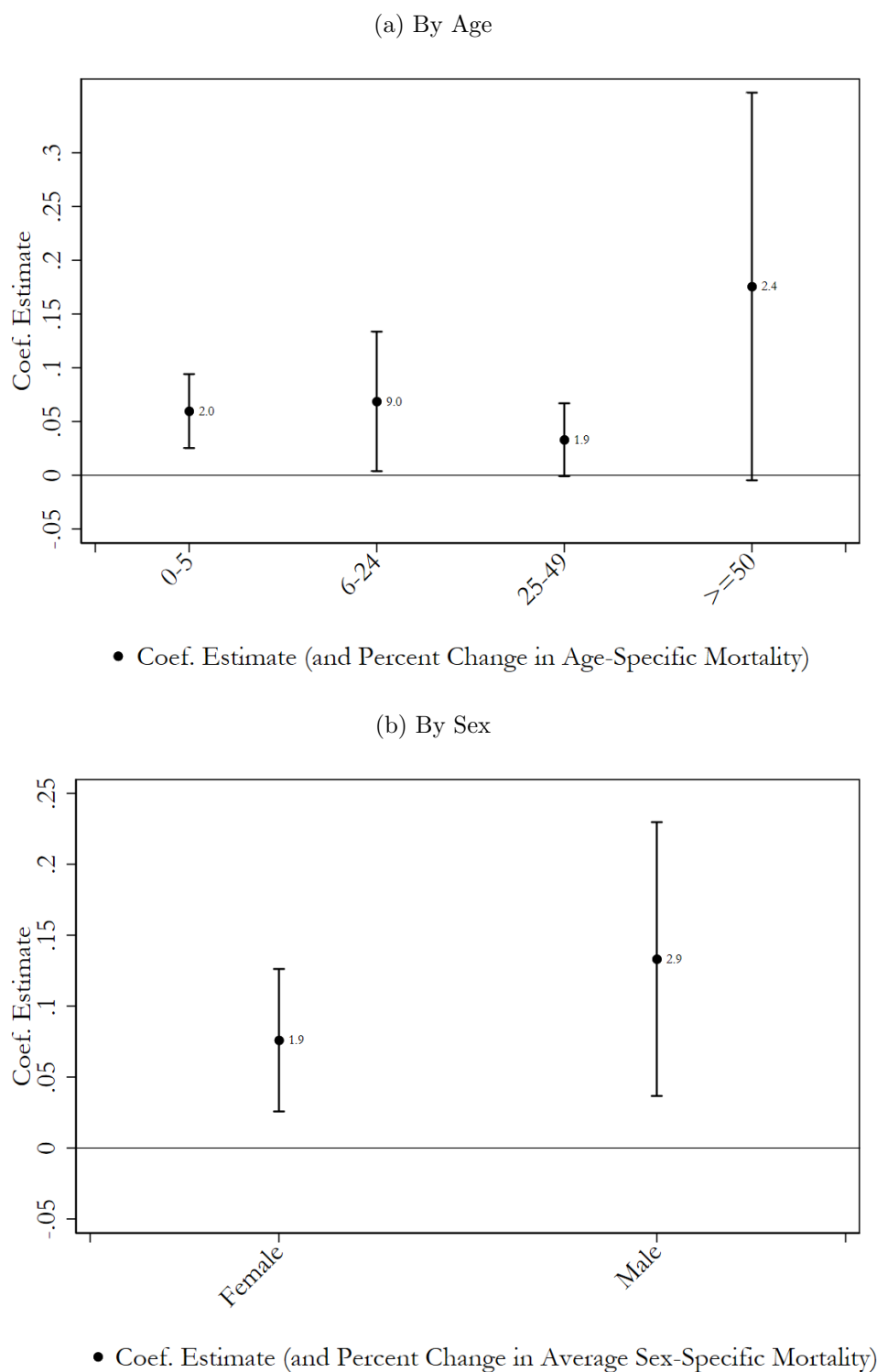
Notes: This figure illustrates the results from a simulation designed to analyze the impact of expected net health benefits on the propensity of insurers to restrict health services, as outlined in the model described in Section 5. In the simulation, we compute optimal decisions by individuals and insurers according to Equations (5.2) and (5.5), varying expected benefits across procedures. The depicted red line represents the threshold beyond which a service becomes subject to restriction. The shaded region delineates the range of values exceeding this threshold. A description of the algorithm is shown in Appendix A.1.

Figure 5: Effect of Restrictions on Access to Medical Care on **Mortality Rates by Cause of Death**



Notes: The Figure shows the estimated effect of supply-side restrictions to access health care on cause-specific mortality rates. Restrictions are measured using the municipality $\times$ insurer $\times$ year tutela rate. The estimates correspond to the instrumental variable specification discussed in Section 6.1. The number to the right of each coefficient indicates the percent change in cause-specific mortality associated with an increase of one tutela per 1,000 enrollees.

Figure 6: Effect of Restrictions on Access to Medical Care on **Mortality Rates by Age and Sex**



Notes: The Figure shows the estimated effect of supply-side restrictions to access to health care on age-specific (Panel (a)) and sex-specific (Panel (b)) mortality rates. The model specification uses complaint rates since the tutela data can not be disaggregated by age or sex. The number to the right of each coefficient indicates the percent change in sex or age-specific mortality associated with a doubling of the complaint rate.

Table 1: Effect of Restrictions on Access to Medical Care on **Utilization of Medical Services** (A): Shift-Share Approach

	Consultations per Enrollee				Procedures per Enrollee			
	OLS (a)	Shift-Share (b)	OLS (c)	Shift-Share (d)	OLS (e)	Shift-Share (f)	OLS (g)	Shift-Share (h)
Tutela Rate	-0.280*** (0.050)	-0.318* (0.189)			-0.183** (0.057)	-0.632*** (0.174)		
Complaint Rate			-0.243*** (0.029)	-0.278*** (0.068)			-0.070** (0.029)	-0.188 (0.115)
Observations	6,121	6,121	5,202	5,202	6,121	6,121	5,202	5,202
Av. Dep. Var.	2.331 (1.093)	2.331 (1.093)	1.990 (1.174)	1.990 (1.174)	1.771 (1.123)	1.771 (1.123)	1.577 (1.160)	1.577 (1.160)
Fixed Effects								
Municipality	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First Stage								
F stat. First Stage		23.369		16.453		23.369		16.453
Effect Size (%)	-12.013	-13.637	-12.220	-13.986	-10.313	-35.689	-4.451	-11.904
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

*** 1 percent ** 5 percent * 10 percent.

Notes: The Table shows the estimated effect of supply-side restrictions to access health care on the utilization of care services. The estimates in this table use municipality-year data that aggregates over insurers. The IV estimates correspond to the shift-share IV discussed in Section 6.2. Each regression includes analytical weights proportional to the number of enrollees in the municipality $\times$ year. We report cluster-robust standard errors clustered by municipality.

Table 2: Effect of Restrictions on Access to Medical Care on **Utilization of Medical Services** (B): Shift-Share Approach

	Hospitalizations per Enrollee				Emergencies per Enrollee			
	OLS (a)	Shift-Share (b)	OLS (c)	Shift-Share (d)	OLS (e)	Shift-Share (f)	OLS (g)	Shift-Share (h)
Tutela Rate	-0.006*** (0.001)	-0.005 (0.003)			-0.016*** (0.004)	-0.047** (0.018)		
Complaint Rate			-0.005*** (0.000)	-0.001 (0.001)			-0.019*** (0.003)	-0.040*** (0.011)
Observations	6,121	6,121	5,202	5,202	6,121	6,121	5,202	5,202
Av. Dep. Var.	0.042 (0.023)	0.042 (0.023)	0.035 (0.025)	0.035 (0.025)	0.130 (0.100)	0.130 (0.100)	0.121 (0.104)	0.121 (0.104)
Fixed Effects								
Municipality	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First Stage								
F stat. First Stage		23.369		16.453		23.369		16.453
Effect Size (%)	-14.070	-10.893	-13.883	-3.812	-12.121	-36.389	-15.443	-32.687
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

*** 1 percent ** 5 percent * 10 percent.

Notes: The Table shows the estimated effect of supply-side restrictions to access health care on the utilization of care services. The estimates in this table use municipality-year data that aggregates over insurers. The IV estimates correspond to the shift-share IV discussed in Section 6.2. Each regression includes analytical weights proportional to the number of enrollees in the municipality $\times$ year. We report cluster-robust standard errors clustered by municipality.

Table 3: Effect of Restrictions on Access to Medical Care on **Mortality Rates**: Three-Way Fixed Effects IV Model

	Dep. Var.: Mortality Rate Subsidized and Contributory Regime							
	OLS (a)	OLS (b)	IV (c)	Shift-Share (d)	OLS (e)	OLS (f)	IV (g)	Shift-Share (h)
Tutela Rate	0.119*** (0.023)	0.082*** (0.014)	0.127*** (0.032)	0.208** (0.069)				
Complaint Rate					0.064** (0.023)	0.050** (0.020)	0.032 (0.021)	0.132*** (0.017)
Observations	44,674	34,435	34,435	6,586	32,729	26,892	26,892	5,496
Fixed Effects								
Municipality $\times$ Year	Yes	Yes	Yes		Yes	Yes	Yes	
Insurer $\times$ Year	Yes	Yes	Yes		Yes	Yes	Yes	
Municipality $\times$ Insurer	Yes	Yes	Yes		Yes	Yes	Yes	
Municipality Year				Yes Yes				Yes Yes
First Stage								
F stat. First Stage			16.17	49.77			19.38	95.54
Effect Size (%)	6.92	4.95	7.62	11.61	8.06	6.28	4.08	14.74
Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes

*** 1 percent ** 5 percent * 10 percent.

Notes: The Table shows the estimated effect of supply-side restrictions to access health care on mortality rates. The IV estimates correspond to the three-way fixed effects IV model discussed in Section 6.1. All rates are defined per 1,000 enrollees, and each regression includes analytical weights proportional to the number of enrollees in the insurer $\times$ municipality $\times$ year. We report cluster-robust standard errors, clustered by municipality and insurer.

Table 4: Effect of Restrictions on Access to Medical Care on **Mortality Rates**: Three-Way Fixed Effects Placebo and Dynamics Model

	Dep. Var.: Mortality Rate Subsidized and Contributory Regime							
	OLS (a)	OLS (b)	OLS (c)	OLS (d)	OLS (e)	OLS (f)	OLS (g)	OLS (h)
Tutela Rate (t+1)	-0.0021 (0.0085)	-0.0053 (0.0115)						
Tutela Rate (t-1)			0.0115 (0.0127)	0.0131 (0.0085)				
Complaint Rate (t+1)					0.0085 (0.0161)	0.0117 (0.0130)		
Complaint Rate (t-1)							-0.0129 (0.0059)	-0.0036 (0.0036)
Observations	34,728	26,894	34,424	34,424	21,008	18,865	21,003	21,003
Fixed Effects								
Municipality $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurer $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality $\times$ Insurer	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Effect Size (%)	-0.06	-0.14	0.30	0.35	0.21	0.29	-0.32	-0.09
Controls	No	Yes	No	Yes	No	Yes	No	Yes

*** 1 percent ** 5 percent * 10 percent.

Notes: The Table shows the estimated effect of supply-side restrictions to access health care on mortality rates, using either lagged or future restrictions instead of contemporary ones. The estimates correspond to the three-way fixed effects model discussed in Section 6.1. All rates are defined per 1,000 enrollees, and each regression includes analytical weights proportional to the number of enrollees in the insurer $\times$ municipality $\times$ year. We report cluster-robust standard errors, clustered by municipality and insurer.

Table 5: Dynamic Selection: Restrictions on Access to Medical Care and **Probability of Changing Insurers**

	Dep. Var.: =1 if insurer changed between t and $t-1$, =0 otherwise					
	Subsidized and Contributory	Subsidized and Contributory	Contributory	Contributory	Subsidized	Subsidized
Tutela Rate (t-1)	0.00039 (0.00022)		0.00003 (0.00022)		0.00092 (0.00075)	
Complaint Rate (t-1)		0.00036 (0.00019)		0.00010 (0.00023)		0.00088 (0.00068)
Observations	163,525,077	131,963,216	89,968,529	72,008,004	70,177,496	56,632,724
Summary Stats.						
Share Insurer Changed	0.14 (0.35)	0.14 (0.35)	0.14 (0.34)	0.14 (0.34)	0.14 (0.35)	0.14 (0.35)
Fixed Effects						
Enrollee	Yes	Yes	Yes	Yes	Yes	Yes
Init. Insurer $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Init. Municipality $\times$ Init. Insurer	Yes	Yes	Yes	Yes	Yes	Yes

*** 1 percent ** 5 percent * 10 percent.

Notes: The Table shows the estimated effect of supply-side restrictions on the probability of changing insurers. The dependent variable is an indicator variable equal to one if the individual changed insurer between year t and $t - 1$. The regressor of interest is the tutela or complaint rate at year $t - 1$. Each regression includes individual fixed effects and initial insurer $\times$ year and initial municipality $\times$ year flexible trends, whereby “initial” we mean the insurer or municipality of the individual when she first appeared in the data. We report cluster-robust standard errors clustered by municipality and insurer.

Table 6: Dynamic Selection: Restrictions on Access to Medical Care and **Probability of Switching Insurers, by Sex and Age**

	Dep. Var.: =1 if insurer changed between t and $t-1$, =0 otherwise					
	Female ages 0-4	Female ages 5-18	Female ages 19-44	Female ages 45-59	Female ages 60-69	Female ages ≥ 70
Tutela Rate (t-1)	0.00060 (0.00035)	0.00038 (0.00027)	0.00044 (0.00024)	0.00042 (0.00023)	0.00034 (0.00020)	0.00022 (0.00017)
Observations	3,646,451	16,479,870	37,954,156	17,316,501	7,067,446	7,340,276
Summary Stats.						
Share Insurer Changed	0.15 (0.36)	0.13 (0.34)	0.17 (0.38)	0.12 (0.32)	0.09 (0.28)	0.07 (0.25)
Fixed Effects						
Enrollee	Yes	Yes	Yes	Yes	Yes	Yes
Init. Insurer $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Init. Municipality $\times$ Init. Insurer	Yes	Yes	Yes	Yes	Yes	Yes

	Dep. Var.: =1 if insurer changed between t and $t-1$, =0 otherwise					
	Male ages 0-4	Male ages 5-18	Male ages 19-44	Male ages 45-59	Male ages 60-69	Male ages ≥ 70
Tutela Rate (t-1)	0.00062 (0.00028)	0.00035 (0.00045)	0.00037 (0.00039)	0.00043 (0.00026)	0.00027 (0.00018)	0.00020 (0.00015)
Observations	3,711,203	15,993,053	27,538,410	12,498,595	5,036,638	4,853,376
Summary Stats.						
Share Insurer Changed	0.15 (0.36)	0.13 (0.33)	0.18 (0.38)	0.14 (0.35)	0.11 (0.31)	0.08 (0.26)
Fixed Effects						
Enrollee	Yes	Yes	Yes	Yes	Yes	Yes
Init. Insurer $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Init. Municipality $\times$ Init. Insurer	Yes	Yes	Yes	Yes	Yes	Yes

*** 1 percent ** 5 percent * 10 percent.

Notes: The Table shows the estimated effect of supply-side restrictions on the probability of changing insurers, dividing the sample by sex and age groups. The dependent variable is an indicator variable equal to one if the individual changed insurer between year t and $t - 1$. The regressor of interest is the tutela or complaint rate at year $t - 1$. Each regression includes individual fixed effects and initial insurer $\times$ year and initial municipality $\times$ year flexible trends, whereby “initial” we mean the insurer or municipality of the individual when she first appeared in the data. We report cluster-robust standard errors clustered by municipality and insurer.

Table 7: Effect of Restrictions on Access to Medical Care on **Mortality Rates by Regime**: Three-Way Fixed Effects IV Model

	Contributory Regime				Subsidized Regime			
	OLS (a)	IV (b)	OLS (c)	IV (d)	OLS (e)	IV (f)	OLS (g)	IV (h)
Tutela Rate	0.055*** (0.011)	0.117** (0.040)			0.166*** (0.034)	0.414** (0.126)		
Complaint Rate			0.049* (0.028)	0.028 (0.021)			0.033** (0.013)	0.370 (0.237)
Observations	17,573	17,573	12,368	12,368	14,251	14,251	11,899	11,899
Fixed Effects								
Municipality $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Insurer $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality $\times$ Insurer	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First Stage								
F stat. First Stage		16.76		16.81		27.82		7.22
Effect Size (%)	4.91	10.40	10.50	5.99	5.27	13.10	1.42	15.95
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

*** 1 percent ** 5 percent * 10 percent.

Notes: The Table shows the estimated effect of supply-side restrictions to access health care on mortality rates, conditional on the regime. The IV estimates correspond to the three-way fixed effects IV model discussed in Section 6.1. All rates are defined per 1,000 enrollees, and each regression includes analytical weights proportional to the number of enrollees in the insurer $\times$ municipality $\times$ year. We report cluster-robust standard errors, clustered by municipality and insurer.

Table 8: Effect of Restrictions on Access to Medical Care
on Mortality Rates by Charlson Index: Three-Way
Fixed Effects IV Model

	Dep. Var.: Mortality Rate Subsidized and Contributory Regime			
	OLS (a)	IV (b)	OLS (c)	IV (d)
Charlson Index ≤ 2				
Tutela Rate	0.024** (0.010)	0.057** (0.017)		
Complaint Rate			0.023*** (0.005)	0.028* (0.017)
Percent Change (%)	3.42	8.16	3.26	4.05
Charlson Index $\in [3, 5]$				
Tutela Rate	0.036** (0.016)	0.068*** (0.014)		
Complaint Rate			0.042*** (0.007)	0.024** (0.011)
Percent Change (%)	2.40	4.66	2.63	1.55
Charlson Index $\in [6, 8]$				
Tutela Rate	0.030** (0.014)	0.074** (0.022)		
Complaint Rate			0.038*** (0.007)	0.021 (0.014)
Percent Change (%)	2.11	5.18	2.44	1.40
Charlson Index ≥ 9				
Tutela Rate	0.003* (0.002)	0.013 (0.010)		
Complaint Rate			0.005** (0.002)	0.002 (0.002)
Percent Change (%)	1.48	6.09	2.19	0.83
Observations	43,822	40,602	34,606	31,866
Fixed Effects				
Municipality $\times$ Year	Yes	Yes	Yes	Yes
Insurer $\times$ Year	Yes	Yes	Yes	Yes
Municipality $\times$ Insurer	Yes	Yes	Yes	Yes
First Stage				
F stat. First Stage		24.04		22.88

*** 1 percent ** 5 percent * 10 percent.

Notes: The Table shows the estimated effect of supply-side restrictions to access health care on mortality rates. The IV estimates correspond to the three-way fixed effects IV model discussed in Section 6.1. All rates are defined per 1,000 enrollees, and each regression includes analytical weights proportional to the number of enrollees in the insurer $\times$ municipality $\times$ year. We report cluster-robust standard errors, clustered by municipality and insurer.

A Online Appendix

A.1 Simulation Algorithm for the Model

The algorithm for simulating the relationship between idiosyncratic health improvements and the decision-making criterion is as follows:

Algorithm 1 Simulation of Model

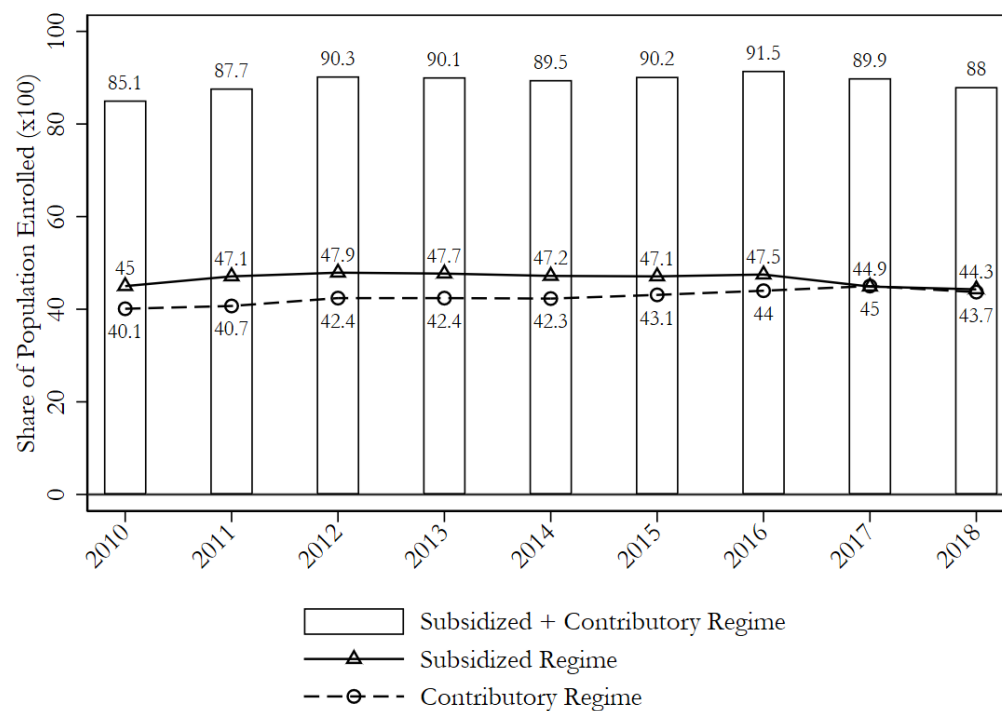
- 1: Initialize parameters: $a = 1$, $\beta = 0.8$, $h_1^j = 10$, $\sigma^2 = 2$, $c = 1$, $\gamma = 1$.
- 2: Define $U(X) = -\exp(-aX)$.
- 3: Determine the number of simulation iterations, set to 10,000.
- 4: Generate 10,000 iid random draws of η^j from $N(0, \sigma^2)$.
- 5: **for** $i = 1$ to the number of simulation iterations **do**
- 6: Generate a random draw of z from $N(3, 1)$.
- 7: Calculate $h_1^{j,x} = h_1^j + z$.
- 8: **for** each draw of η^j **do**
- 9: Calculate $H(\eta_j)$ using the formula:

$$H(\eta_j) = \frac{U(h_1^j) - U(h_1^j - c)}{U(h_1^{j,x} + \eta^j - c) - U(h_1^j - c)}$$

- 10: **end for**
 - 11: Find the fraction of observations, denoted α^j , such that $H(\eta_j) \leq \beta$.
 - 12: Calculate *Cost of procedure / Admin.* $Cost = \frac{\alpha^j}{(1-\alpha^j\beta)}$.
 - 13: **end for**
 - 14: Plot the relation between *Cost of procedure / Admin. Cost* and *Expected net health benefit (%)* $= 100 * (h_1^{j,x} - h_1^j) / h_1^j$.
-

A.2 Appendix Tables and Figures

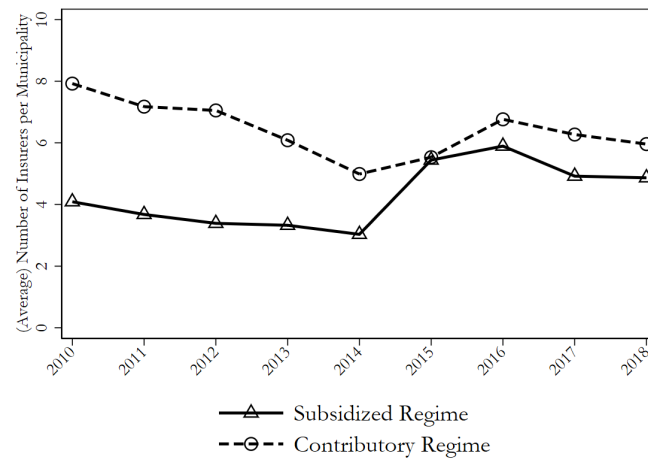
Figure A.1: Coverage of the Health System: Overall and by Regime



Notes: The bars show the share of the the population that is either in the contributory or subsidized regimes by year. The two lines report the share of population that is in the subsidized regime (solid) and contributory regime (dashed) by year.

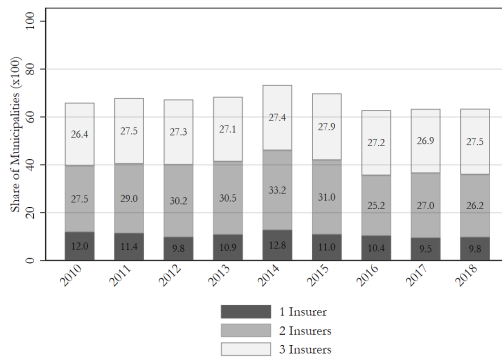
Figure A.2: Competitive Structure of the Health Care System

(a) Insurers per Municipality

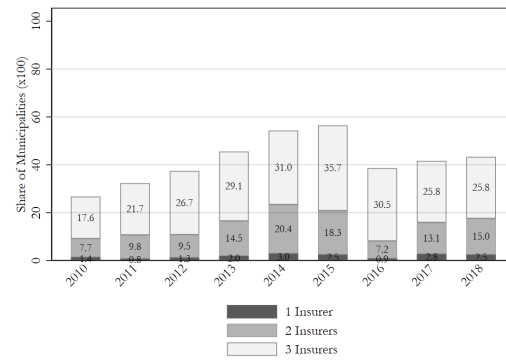


Share of Municipalities with at Most One, Two or Three Insurers

(b) Subsidized Regime



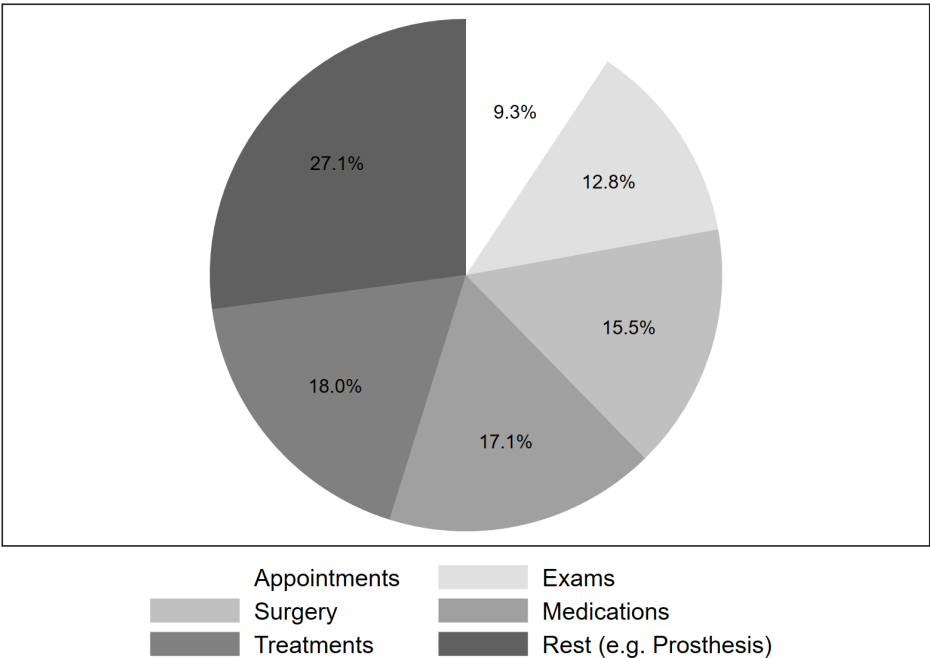
(c) contributory Regime



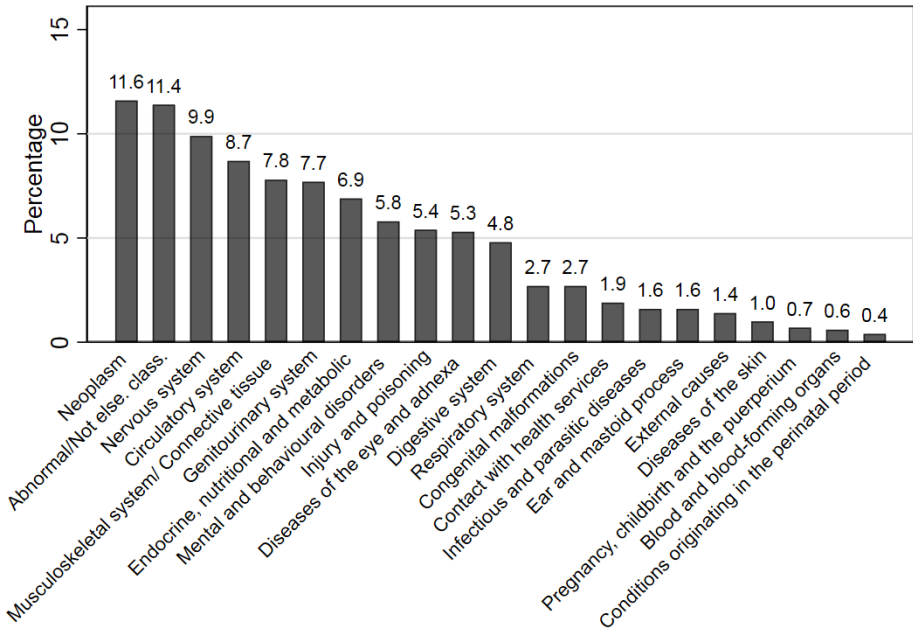
Notes: Panel (a) shows the average number of insurers per municipality and by regime. Panels (b) and (c) show the share ($\times 100$) of municipalities that have at most 1, 2, or 3 insurers by regime.

Figure A.3: Type of Treatment Requested and Diagnoses Associated with the Tutelas Invoking the Right-to-Health

(a) Type of Treatment Requested in Tutelas Invoking the Right-to-Health



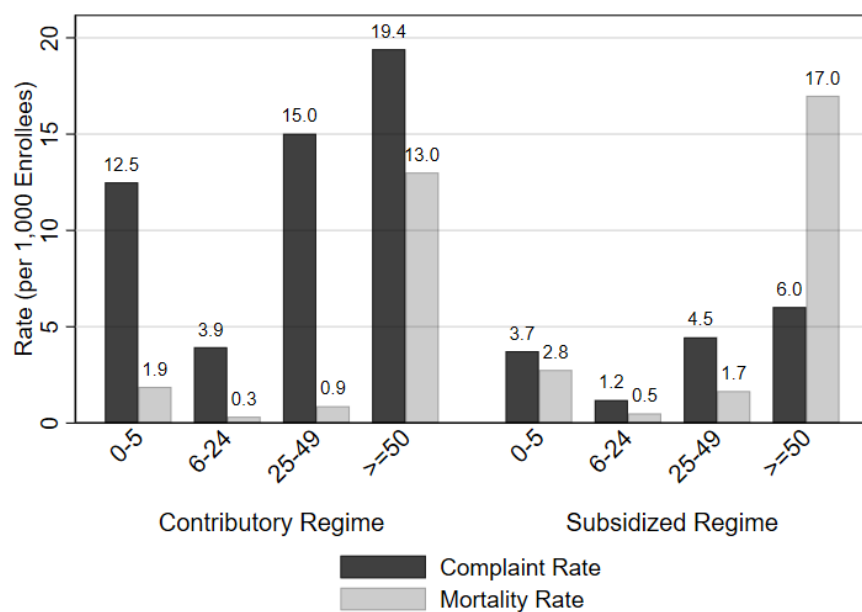
(b) Diagnoses Associated with the Tutelas in 2015



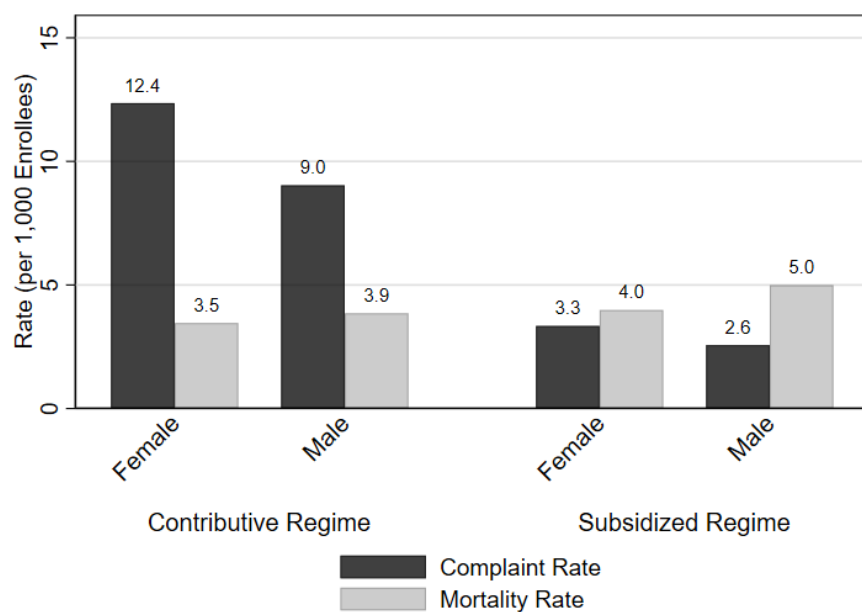
Source for Panel (a): authors calculation based on Defensoría del Pueblo (2004, 2007, 2009, 2010, 2011, 2012, 2013, 2015). Source for Panel (b): Minsalud (2015). The numbers reported in both panels include tutelas made against other actors in the system different from insurers (i.e. IPSs, Secretary of Health and Social Protection, local authorities).

Figure A.4: Complaint and Mortality Rates by Age and Sex Conditional on Regime

(a) By Age



(b) By Sex



Notes: Panel (a) reports the average complaint and mortality rates by age. Panel (b) reports the average complaint and mortality rates by sex. The divisions correspond to the people affected by the complaint, not those that file it

Table A.1: Complaints to the National Health Superintendence Between 2013 and 2017

	Overall		Contributory Regime		Subsidized Regime	
	Number	Share	Number	Share	Number	Share
Restriction of Access to Care Services*	737,785.0	44.1	565,273.0	47.7	119,172.0	32.1
Delays of Authorizations**	305,508.0	18.3	185,351.0	15.6	101,967.0	27.5
Problems with Affiliation	144,362.0	8.6	97,255.0	8.2	41,462.0	11.2
Restriction to Change/Choose Insurer/Provider	84,311.0	5.0	46,828.0	4.0	35,716.0	9.6
Denials of Authorizations**	75,309.0	4.5	51,778.0	4.4	18,445.0	5.0

Notes: The numbers in the overall column include complaints made against other actors in the system different from insurers (i.e. IPSs, Secretary of Health and Social Protection, local authorities). The numbers by regime include only complaints made against insurers. *Includes lack of opportunity to get an appointment with a care specialist, lack of opportunity to program a surgery, lack of opportunity to program an exam, and lack of opportunity to get a medication amongst others. **Includes authorizations for appointments with care specialists, surgeries, exams, and medications.

Table A.2: Dynamic Selection: Restrictions on Access to Medical Care and Probability of Changing the Municipality of Residence

	Dep. Var.: =1 if municipality changed between t and $t-1$, =0 otherwise					
	Subsidized and Contributory	Subsidized and Contributory	Contributory	Contributory	Subsidized	Subsidized
Tutela Rate (t-1)	0.00009 (0.00009)		0.00001 (0.00012)		0.00020* (0.00008)	
Complaint Rate (t-1)		0.00012 (0.00011)		0.00006 (0.00015)		0.00025 (0.00017)
Observations	163,453,484	131,891,619	89,939,941	71,979,416	70,139,094	56,594,318
Summary Stats.						
Share Mpio. Changed	0.06 (0.23)	0.06 (0.23)	0.07 (0.25)	0.07 (0.25)	0.04 (0.20)	0.04 (0.20)
Fixed Effects						
Enrollee	Yes	Yes	Yes	Yes	Yes	Yes
Init. Insurer $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Init. Municipality $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes

*** 1 percent ** 5 percent * 10 percent.

Notes: The Table shows the estimated effect of supply-side restrictions on the probability of migrating to another municipality. The dependent variable is an indicator variable equal to one if the individual changed municipality between year t and $t - 1$. The regressor of interest is the tutela or complaint rate at year $t - 1$. Each regression includes individual fixed effects and initial insurer $\times$ year and initial municipality $\times$ year flexible trends, whereby “initial” we mean the insurer or municipality of the individual when she first appeared in the data. We report cluster-robust standard errors clustered by municipality and insurer.